

Fractional ED3 Instantons on $\mathbb{C}^3/\mathbb{Z}_3$ Orientifolds

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Abstract

This note computes the worldsheet CFT data needed for the one-loop normalization K_0 of a fractional ED3 instanton on the orientifold of $\mathbb{C}^3/\mathbb{Z}_3$ with an O7-plane on the exceptional divisor. The phrase “D-instanton” is used in the four-dimensional sense: an instanton is localized in $\mathbb{R}^{1,3}$, but it may wrap an internal cycle. At the orbifold point, the wrapped ED3 is represented by a fractional boundary condition. The A_2 -theta DN annulus and the orbifold-averaged one-node $O(1)$ folded Mobius trace are evaluated explicitly. The determinant calculation is complete for this folded branch. It gives a standalone APS prefactor only when the D7 imbalance seen by the instanton and the projected non-universal zero-mode space both vanish. Under those two conditions

$$\log K_0 = -2.8226016706417078, \quad K_0 = 0.05945106922997144.$$

In the ordinary local D7 realization with $N_0 = N_1 = N_2 = 8$, charged ED3-D7 zero modes survive, so the determinant factor is known but is not by itself a standalone superpotential prefactor.

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1 Purpose and Conventions

The goal is to compute the one-loop factor K_0 entering the superpotential contribution of an $O(1)$ fractional ED3 instanton. A numerical value of K_0 requires local fractional open-string multiplicities, orientifold eigenvalues, and zero-mode data for one specified orientifold BCFT. Notational normalizations are stated explicitly. Physical signs, lifts, and crosscap coefficients are not free choices: in the complete BCFT they are fixed by modular invariance, tadpole factorization, and the orientifold bootstrap. This note computes the annulus contribution and the orbifold-averaged one-node folded Mobius contribution explicitly. For the branch defined by these data there is no additional twisted Mobius term to tune. If the D7 vector produces a nonzero twisted annulus tadpole, then this branch is not a complete tadpole-cancelling BCFT for a standalone K_0 integral.

In this note K_0 means the standalone numerical one-loop prefactor in the Alexandrov–Pioline–Sen formula after the universal translation and Goldstino zero modes have been treated. If additional $L_0 = 0$ ED3-D7 states survive, the assumptions of that standalone K_0 formula are not

satisfied; the determinant factor can still be computed, but it is not by itself the superpotential prefactor.

Two pieces of terminology will be kept separate throughout the note. The *one-node folded branch* refers to the orientifold action on the fractional ED3 and its ED3-ED3 Mobius trace: the instanton label is fixed by the orientifold, and Ω_σ pairs charge-conjugate internal oscillators in the trace. The *ordinary symmetric D7 model* refers instead to a space-filling D7 background, namely $N_0 = N_1 = N_2 = 8$. The first phrase fixes the neutral instanton Mobius sector; the second phrase fixes the charged ED3-D7 sectors. They are compatible data, but they answer different questions.

Terminology. The following terms are used throughout the note.

APS Alexandrov–Pioline–Sen, whose D-instanton normalization formula is recalled in (183).

BCFT Boundary conformal field theory. Here it means the local orbifold worldsheet CFT together with its boundary states, crosscap state, GSO projection, Chan–Paton action, and orientifold lift.

parent CFT The unorbifolded free-field CFT before the $\mathbb{Z}_3$ projection. Fractional open-string spectra are obtained by taking eigenspaces of parent-CFT open-string Hilbert spaces.

fractional label or node

An element of $\mathbb{Z}/3\mathbb{Z}$ labelling an irreducible equivariant Chan–Paton character. It is not a spacetime index or a coordinate index.

twined trace An open-string trace with an insertion of g^ℓ , where g is the orbifold generator. Fourier transforming the three twined traces gives the three fractional charge sectors.

DD and DN annuli

DD means both annulus boundaries are on the ED3 instanton. DN means one boundary is on the ED3 and the other is on the D7 stack; the four non-compact spacetime directions then have Dirichlet–Neumann boundary conditions.

Mobius The Mobius strip contribution. The accent is omitted in formulas and labels; it always refers to the unoriented open-string trace with an orientifold insertion.

folded trace A Mobius trace in which the orientifold maps a state to its charge conjugate. The trace is then over charge-conjugate oscillator pairs, not over a single ordinary $U(1)$ -charge sector.

determinant integrand

The open-channel trace after the $L_0 = 0$ zero-mode representatives have been removed, but before the t -integral has been performed.

determinant factor

The finite exponential obtained from the determinant integrands. It is a standalone superpotential prefactor only if no non-universal zero modes remain.

standalone K_0

The numerical APS one-loop prefactor after removing only the four translations and two universal Goldstinos. If charged or extra neutral $L_0 = 0$ modes survive, this standalone object is not the correct superpotential prefactor.

projected zero-mode space

The $L_0 = 0$ vertex space after the GSO, orientifold, and Chan–Paton projections have all been imposed.

twisted tadpole compatibility

Absence of an uncanceled twisted closed-string massless exchange in the small- t endpoint of the annulus-plus-Mobius integrand.

massless row One of the three massless internal $\mathcal{N} = 2$ character rows $\lambda \in \{\text{vac}, +, -\}$. A row is a character contribution to a trace; it is not automatically a list of $L_0 = 0$ vertex representatives.

The result proved below can be stated at the outset. Let $a \in \mathbb{Z}/3\mathbb{Z}$ be the fractional ED3 label, and let N_m be the D7 boundary-state coefficient of fractional label $m \in \mathbb{Z}/3\mathbb{Z}$. Define the D7 imbalance seen by the instanton by

$$\mathcal{D}_a = 2N_a - N_{a-1} - N_{a+1}. \quad (1)$$

The determinant BCFT data in the folded one-node branch are

$$Z_{A,\text{DD}}^{\text{det}}(t) = 0, \quad Z_{A,\text{DN}}^{(a),\text{det}}(t) = \frac{3}{4} \mathcal{D}_a \left(\Theta_{A_2}(e^{-2\pi t}) - 1 \right), \quad Z_M^{\text{det}}(t) = C_M^{\text{fold}}(t). \quad (2)$$

Here Θ_{A_2} is the A_2 root-lattice theta function defined in (207), and C_M^{fold} is the folded Mobius integrand defined in (253). These data define a standalone APS K_0 exactly when

$$\mathcal{D}_a = 0, \quad \mathcal{Z}_{\text{extra}}^{\text{proj}}(a) = 0, \quad (3)$$

where $\mathcal{Z}_{\text{extra}}^{\text{proj}}(a)$ is the projected $L_0 = 0$ zero-mode space after removing the four translations and the two universal Goldstinos. When (3) holds, the value is

$$\begin{aligned} \log K_0(a) &= -2.82260167064170776861041806382945297770, \\ K_0(a) &= 0.05945106922997143616905682111809587940. \end{aligned} \quad (4)$$

The derivation proceeds as follows. First the note fixes the orbifold, fractional-brane, orientifold, and tadpole conventions. It then derives the fractional open-string multiplicities and the charged zero-mode test. The determinant computation comes next: the DN annulus resums to the A_2 -theta expression, the ED3-ED3 Mobius strip is evaluated as a folded trace, and the small- t endpoints are compared. The final section records the K_0 status in formulas.

Convention 1 (Meaning of D-instanton). *In the terminology of Alexandrov–Pioline–Sen, a “D-instanton” means any D-brane localized at a point in the non-compact spacetime $\mathbb{R}^{1,3}$, regardless of how it extends in the internal Calabi-Yau. Thus a Euclidean D3-brane wrapping an internal four-cycle is a D-instanton for the four-dimensional theory.*

In this note, the fractional ED3 is the orbifold CFT boundary condition whose resolved-space interpretation is an ED3 wrapping the exceptional divisor. This is not the same as an ordinary ten-dimensional D(−1)-brane. Whenever the phrase “fractional instanton” appears, it refers to the former object.

2 Geometry of $\mathbb{C}^3/\mathbb{Z}_3$

Let (z_1, z_2, z_3) be complex coordinates on $\mathbb{C}^3$. The orbifold group is generated by

$$g : (z_1, z_2, z_3) \mapsto (\omega z_1, \omega z_2, \omega z_3), \quad \omega = e^{2\pi i/3}. \quad (5)$$

The holomorphic three-form $\Omega_3 = dz_1 \wedge dz_2 \wedge dz_3$ is invariant because $\omega^3 = 1$. The unique fixed point of the orbifold action on $\mathbb{C}^3$ is the origin.

The crepant resolution is

$$X = \text{Tot}(\mathcal{O}_{\mathbb{CP}^2}(-3)). \quad (6)$$

We write a point of X as $[z_1, z_2, z_3; p]$, where $[z_1, z_2, z_3]$ are homogeneous coordinates on $\mathbb{CP}^2$ and p is the fiber coordinate of $\mathcal{O}(-3)$. The equivalence relation is

$$(z_1, z_2, z_3; p) \sim (\lambda z_1, \lambda z_2, \lambda z_3; \lambda^{-3} p), \quad \lambda \in \mathbb{C}^\times. \quad (7)$$

The exceptional divisor is

$$E = \{p = 0\} \simeq \mathbb{CP}^2. \quad (8)$$

The fractional ED3 considered here is the orbifold-point representative of an ED3 wrapping E .

Assumption 1 (Freed-Witten datum). *The resolved-space ED3 on $E \simeq \mathbb{CP}^2$ is only meaningful after choosing a worldvolume bundle and B-field. Since $w_2(T\mathbb{CP}^2)$ is the hyperplane class modulo 2, the gauge-invariant flux must obey the Freed-Witten condition*

$$\mathcal{F} + \frac{1}{2}H \in H^2(E, \mathbb{Z}), \quad H = c_1(\mathcal{O}_{\mathbb{CP}^2}(1)), \quad (9)$$

where $\mathcal{F}$ denotes the combination of worldvolume flux and pulled-back B-field in the normalization in which integral line bundles have integral first Chern class. In this note the fractional boundary label is taken to include this discrete datum. Changing it changes the RR charge vector and can change the orientifold parity and zero-mode spectrum.

3 Fractional Boundary Conditions

This section uses the orbifold boundary-state construction reviewed in the string book. Let B be a boundary condition of the parent CFT that is invariant under the $\mathbb{Z}_3$ action. Let $|B, g^\ell\rangle$ denote the boundary state with a g^ℓ defect ending on the boundary.

Definition 1 (Fractional boundary condition). *For $m \in \{0, 1, 2\}$, define*

$$|\tilde{B}_m\rangle = \frac{1}{\sqrt{3}} \sum_{\ell=0}^2 \omega^{m\ell} |B, g^\ell\rangle. \quad (10)$$

The label m is a vertex label, not a transverse coordinate index. It records the character $\chi_m(g) = \omega^m$ of the equivariant Chan-Paton representation.

The open-string Hilbert space between two fractional boundary conditions is not obtained by summing over closed-string twisted sectors in the open channel. It is an eigenspace of the parent open-string Hilbert space:

$$\mathcal{H}_{\tilde{B}_m \tilde{B}_{m'}} \simeq (\mathcal{H}_{BB})_{\omega^{m-m'}}. \quad (11)$$

Here $(\mathcal{H}_{BB})_{\omega^r}$ means the subspace on which the parent orbifold generator g acts with eigenvalue ω^r .

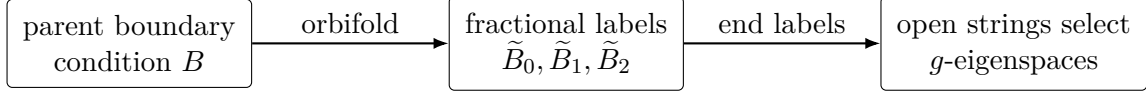


Figure 1: Fractional branes are boundary conditions in the orbifold CFT. Their labels determine which eigenspace of the parent open-string Hilbert space appears. The drawing suppresses oscillator content and only shows the bookkeeping map from boundary labels to projections.

3.1 Fractional ED3 and Fractional D7 Labels

Let $\tilde{I}_a$ denote the fractional ED3 instanton boundary condition, with $a \in \{0, 1, 2\}$. Let $\tilde{S}_m$ denote a fractional space-filling D7 boundary condition, with $m \in \{0, 1, 2\}$. The notation S stands for “space-filling” in $\mathbb{R}^{1,3}$; it does not mean that the internal boundary condition is the same as that of the instanton.

For an oriented string from the instanton to a space-filling brane, define

$$\mathcal{H}_{am}^{IS} := \mathcal{H}_{\tilde{I}_a \tilde{S}_m}^{IS}. \quad (12)$$

If the parent internal boundary conditions are I and S , then

$$\mathcal{H}_{am}^{IS} \simeq (\mathcal{H}_{IS})_{\omega^{a-m}}. \quad (13)$$

The reverse orientation uses the conjugate eigenspace $(\mathcal{H}_{SI})_{\omega^{m-a}}$.

4 Orientifold Involutions

An orientifold action is a worldsheet orientation reversal combined with a target-space involution. For the O7 orientifold used below we write it as

$$\Omega_\sigma = (-1)^{F_L} \Omega \circ \sigma. \quad (14)$$

Here Ω reverses the worldsheet orientation, F_L is the left-moving spacetime fermion number, and σ acts on the target space. Omitting $(-1)^{F_L}$ would define a different orientifold with different RR crosscap signs, so the coefficients C_ℓ below would have to be recomputed.

The orientifold also acts on fractional labels. In a lift for which

$$\Omega_\sigma |B, g^\ell\rangle = \kappa_\ell |B, g^{-\ell}\rangle, \quad \kappa_\ell \in U(1), \quad (15)$$

one has

$$\Omega_\sigma |\tilde{B}_m\rangle = \frac{1}{\sqrt{3}} \sum_{\ell=0}^2 \omega^{m\ell} \kappa_\ell |B, g^{-\ell}\rangle. \quad (16)$$

For the minimal lift $\kappa_\ell = 1$, this gives

$$\Omega_\sigma |\tilde{B}_m\rangle = |\tilde{B}_{-m}\rangle. \quad (17)$$

Thus label 0 can support a single invariant O(1) instanton, while labels 1 and 2 are exchanged and require an invariant linear combination. A nontrivial lift κ_ℓ changes this label map and is part of the crosscap data.

A common restricted class of lifts multiplies the minimal lift by a quantum-symmetry character,

$$\kappa_\ell = \omega^{s\ell}, \quad s \in \{0, 1, 2\}. \quad (18)$$

Then

$$\Omega_\sigma |\tilde{B}_m\rangle = |\tilde{B}_{s-m}\rangle. \quad (19)$$

An invariant one-node $O(1)$ instanton must obey $2a = s \pmod{3}$. Thus, after the bootstrap fixes the lift s , the invariant label is $a = 2s \pmod{3}$. This lift must be fixed before computing the ED3-D7 zero-mode spectrum.

For later use, the three character lifts are:

lift s	label map	one-node invariant label	D7-image constraint
0	$m \mapsto -m$	$a = 0$	$N_1 = N_2$
1	$m \mapsto 1 - m$	$a = 2$	$N_0 = N_1$
2	$m \mapsto 2 - m$	$a = 1$	$N_0 = N_2$

Here N_m is the D7 boundary-state coefficient defined in (22). These equalities only identify orientifold images. They do not replace the RR tadpole equations.

4.1 O7 Involution on the Exceptional Divisor

The involution used in the K_0 computation is

$$\sigma_1 : [z_1, z_2, z_3; p] \mapsto [z_1, z_2, z_3; -p]. \quad (20)$$

Its fixed locus is $E = \{p = 0\} \simeq \mathbb{CP}^2$, so it produces an O7-plane wrapping the exceptional divisor. On the $\mathbb{C}^3$ cover of the orbifold point the same involution is represented by

$$R : (Z^1, Z^2, Z^3) \mapsto (-Z^1, -Z^2, -Z^3). \quad (21)$$

Indeed, the projective equivalence $(z_1, z_2, z_3; p) \sim (-z_1, -z_2, -z_3; -p)$ is the special case $\lambda = -1$ of (7). Thus σ_1 is trivial on the homogeneous base point $[z_1, z_2, z_3]$, but it is not trivial on the cover fields whose oscillators enter the orbifold BCFT. In an open-channel twined trace the oscillator phase associated with Rg^ℓ is therefore $-\omega^\ell$ on Z^i and $-\omega^{-\ell}$ on $\bar{Z}^i$. The discrete datum still to specify is the lift of this geometric action to the fractional Chan–Paton and spin structures; that lift is fixed by the unoriented modular bootstrap.

5 O(1) Instanton Convention and Universal Zero Modes

Assumption 2 (O(1) projection). *The fractional ED3 instanton is invariant under the orientifold, possibly after identifying it with its image fractional label. Its Chan–Paton projection is fixed so that the four translation modes in $\mathbb{R}^{1,3}$ survive and the unwanted gauge ghost zero modes are projected out. This is the O(1) instanton convention.*

Under this assumption the universal zero modes are:

- four bosonic zero modes x^μ , $\mu = 0, 1, 2, 3$, for translations in $\mathbb{R}^{1,3}$;
- two fermionic zero modes θ^α , $\alpha = 1, 2$, associated with the two supercharges broken by the O(1) instanton.

The absence of additional ED3 deformation modes is a rigidity assumption about the fractional instanton boundary condition. If extra neutral or charged zero modes appear in the fractional open-string spectrum, the standalone K_0 formula used in this note is not the applicable object.

6 RR Tadpoles and the Crosscap Vector

Let N_m be the total boundary-state coefficient of fractional D7-branes of label m in the orientifold theory. Thus N_m counts the branes and images that actually couple to RR fields in the quotient convention used for the annulus formula; it is not an additional covering-space multiplicity. The total D7 boundary state is

$$|S\rangle_{\text{RR}} = \sum_{m=0}^2 N_m |\tilde{S}_m\rangle_{\text{RR}} = \frac{1}{\sqrt{3}} \sum_{\ell=0}^2 \left(\sum_{m=0}^2 N_m \omega^{m\ell} \right) |S, g^\ell\rangle_{\text{RR}}. \quad (22)$$

Let the RR crosscap state be written as

$$|C\rangle_{\text{RR}} = \sum_{\ell=0}^2 C_\ell |C, g^\ell\rangle_{\text{RR}}, \quad (23)$$

where C_ℓ are crosscap coefficients. The symbols $|S, g^\ell\rangle_{\text{RR}}$ and $|C, g^\ell\rangle_{\text{RR}}$ denote RR Ishibashi-state components in the g^ℓ sector; their normalization must be fixed before comparing C_ℓ with the D7 coefficients.

The local RR tadpole cancellation condition is the sector-by-sector equation

$$\frac{1}{\sqrt{3}} \sum_{m=0}^2 N_m \omega^{m\ell} + C_\ell = 0, \quad \ell = 0, 1, 2, \quad (24)$$

after all states are expressed in the same Ishibashi basis.

Equivalently, once the crosscap coefficients C_ℓ are known in this basis, the D7 vector is fixed by the inverse discrete Fourier transform

$$N_m = -\frac{1}{\sqrt{3}} \sum_{\ell=0}^2 C_\ell \omega^{-m\ell}, \quad m = 0, 1, 2. \quad (25)$$

This formula is exact in the stated boundary-state normalization. A proposed crosscap vector is therefore admissible only if the right hand side gives nonnegative integers compatible with the orientifold image constraints for the lift s .

Assumption 3 (Symmetric D7 distribution). *When the crosscap has no uncanceled twisted RR charge in the stated normalization, a symmetric distribution*

$$N_0 = N_1 = N_2 = N \quad (26)$$

cancels the boundary contribution in the $\ell = 1, 2$ twisted sectors because $1 + \omega + \omega^2 = 0$. If the untwisted $O7$ charge is -8 in bulk D7 units, then in the normalization of (24) this means $C_0 = -8\sqrt{3}$ and $N = 8$ gives 24 fractional D7 boundary-state units, corresponding to 8 bulk D7-branes.

The symmetric D7 assumption is not a replacement for a crosscap computation. If C_1 or C_2 is nonzero, then the D7 multiplicities must solve the full vector equation (24).

6.1 Crosscap Vector from Factorization

The coefficients C_ℓ are fixed by factorizing unoriented one-loop diagrams into the closed-string channel.

Let

$$B_m^\ell = \frac{\omega^{m\ell}}{\sqrt{3}}, \quad Q_S^\ell = \sum_{m=0}^2 N_m B_m^\ell = \frac{1}{\sqrt{3}} \sum_{m=0}^2 N_m \omega^{m\ell}. \quad (27)$$

Here B_m^ℓ is the coefficient of the RR Ishibashi block labelled by g^ℓ in the fractional D7 boundary state $\tilde{S}_m$, and Q_S^ℓ is the total D7 charge coefficient in that block. In the notation of (24), Q_S^ℓ is the first term.

The following equations display only the coefficient of a fixed massless RR Ishibashi block. The common closed-string propagator, ghost insertion, and oscillator character in that block are denoted by $\mathcal{I}_\ell(\ell_c)$, where ℓ_c is the closed-channel Schwinger parameter. Thus no oscillator or normalization factor is being silently discarded; all such factors are contained in the same object $\mathcal{I}_\ell$ on the annulus, Mobius, and Klein bottle sides. With this convention, the RR tadpole part of the tree-channel D7 annulus, D7 Mobius strip, and Klein bottle has the form

$$\tilde{\mathcal{A}}_{SS}^{\text{RR}} = \sum_{\ell=0}^2 Q_S^\ell Q_S^{-\ell} \mathcal{I}_\ell, \quad (28)$$

$$\tilde{\mathcal{M}}_S^{\text{RR}} = 2 \sum_{\ell=0}^2 Q_S^\ell C_{-\ell} \mathcal{I}_\ell, \quad (29)$$

$$\tilde{\mathcal{K}}^{\text{RR}} = \sum_{\ell=0}^2 C_\ell C_{-\ell} \mathcal{I}_\ell. \quad (30)$$

The tilde means that the expression has been modular transformed to the closed-string channel. For real D7 multiplicities, $Q_S^{-\ell} = (Q_S^\ell)^*$. A consistent crosscap has the analogous reality condition $C_{-\ell} = C_\ell^*$ in a conjugation-compatible Ishibashi basis. Therefore the sum of the three tree-channel RR tadpole contributions factorizes as

$$\tilde{\mathcal{A}}_{SS}^{\text{RR}} + \tilde{\mathcal{M}}_S^{\text{RR}} + \tilde{\mathcal{K}}^{\text{RR}} = \sum_{\ell=0}^2 (Q_S^\ell + C_\ell) (Q_S^{-\ell} + C_{-\ell}) \mathcal{I}_\ell. \quad (31)$$

Equation (24) is exactly the condition that each factor $Q_S^\ell + C_\ell$ vanishes.

In practice the crosscap vector is obtained in two stages. First, compute the direct-channel Klein bottle trace with orientifold insertions

$$\mathcal{K}^{\text{RR}}(t) = \frac{1}{2 \cdot 3} \sum_{r=0}^2 \text{Tr}_{\mathcal{H}_{\text{closed}}^{\text{RR}}} \left(\Omega_\sigma g^r P_{\text{GSO}}^{\text{closed}} e^{-2\pi t(L_0 + \bar{L}_0 - c/12)} \right), \quad (32)$$

where $P_{\text{GSO}}^{\text{closed}}$ is the closed-string GSO projector and c is the total worldsheet central charge of the traced sector. After the appropriate Klein-bottle modular transformation, (32) fixes the products $C_\ell C_{-\ell}$ in (30). This fixes magnitudes and relative support in twisted sectors, but by itself it may not fix all phases of C_ℓ .

Second, compute a Mobius amplitude with a probe fractional D7 boundary state whose coefficients B_m^ℓ are already known from (27). In the closed channel it must have the coefficient structure

$$\tilde{\mathcal{M}}_m^{\text{RR}} = 2 \sum_{\ell=0}^2 B_m^\ell C_{-\ell} \mathcal{I}_\ell. \quad (33)$$

This comparison fixes the phases of C_ℓ relative to the fractional D7 boundary-state basis and also fixes the lift phases κ_ℓ in (15). This is the step hidden by the informal phrase “O7 charge multiplier” in rough calculations. The O7 charge is a closed-channel crosscap coefficient; it is not a scalar multiplying the open-channel Möbius character.

There are three useful consistency checks on the result. First,

$$C_{-\ell} = C_\ell^* \quad (34)$$

in a basis where the g^ℓ and $g^{-\ell}$ Ishibashi blocks are conjugate. Second, the inverse transform (25) must give nonnegative integers N_m . Third, for a character lift $m \mapsto s - m$, the solution must obey the image constraint $N_m = N_{s-m}$.

Proposition 1 (Untwisted charge already forces local D7 charge). *Assume that the D7 multiplicities obtained from (25) are nonnegative integers. Then*

$$\sum_{m=0}^2 N_m = -\sqrt{3} C_0. \quad (35)$$

In particular, if the involution-I O7⁻ crosscap has $C_0 = -8\sqrt{3}$ in the normalization of (24), then every tadpole solution has total local D7 multiplicity

$$\sum_{m=0}^2 N_m = 24. \quad (36)$$

Proof. Sum (25) over m . Since $\sum_{m=0}^2 \omega^{-m\ell} = 3\delta_{\ell,0}$, only the untwisted coefficient C_0 remains. $\square$

The consequence for a one-node invariant instanton is stated precisely in Corollary 1, after the local ED3-D7 zero-mode test has been derived.

7 Theta Functions and Characters

We use the theta-function convention

$$q = e^{2\pi i \tau}, \quad \vartheta_{\alpha\beta}(\tau) = \sum_{n \in \mathbb{Z}} (-1)^{\beta n} q^{(n+\alpha/2)^2/2}, \quad \alpha, \beta \in \{0, 1\}. \quad (37)$$

Thus ϑ_{00} , ϑ_{01} , and ϑ_{10} are the usual θ_3 , θ_4 , and θ_2 at zero argument, while $\vartheta_{11}(\tau) = 0$. When a characteristic such as $\alpha + Q$ appears below, it is reduced modulo 2 with this convention. The Dedekind eta function is

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n). \quad (38)$$

The internal Calabi-Yau SCFT has central charge $c = 9$ and an extended $\mathcal{N} = 2$ superconformal algebra. Massive representations are labelled by (h, Q) , where h is the conformal weight and $Q \in \{0, 1\}$ is the charge label used in the extended-character convention. For a massive representation, the character components are

$$g_{\alpha\beta}^{(h,Q)}(\tau) = (-1)^{\beta Q} \frac{q^{h-(1+Q)/4}}{\eta(\tau)^3} \vartheta_{\alpha\beta}(\tau) \vartheta_{\alpha+Q,0}(2\tau). \quad (39)$$

Massless representations are denoted (vac), (+), and (−). The vacuum representation contains the $SL(2, \mathbb{R})$ invariant vacuum.

8 Annulus Contributions

In this section $Z_{A,DD}$ means the annulus with both boundaries on the fractional ED3, and $Z_{A,DN}$ means the annulus with one boundary on the ED3 and one boundary on the D7 stack. The letters DD and DN refer to the four non-compact spacetime directions; the internal $\mathbb{C}^3/\mathbb{Z}_3$ data are carried by the fractional labels and character multiplicities.

8.1 Both Boundaries on the Fractional Instanton

Let $Z_{A,DD}$ be the annulus contribution with both boundaries on the fractional instanton. For each representation of the extended $\mathcal{N} = 2$ algebra, the universal free-field identity gives

$$Z_{A,DD}^{(r)} = 0. \quad (40)$$

This is exact under the assumptions of the reference: no additional zero modes beyond the universal $O(1)$ modes and a boundary condition preserving the relevant extended algebra. The identity is not a statement that the instanton has no zero modes; it is a statement about the massive non-zero-mode annulus contribution after the supersymmetric spin-structure sum.

8.2 One Boundary on the Instanton and One on a Space-filling Brane

The universal open-string block imported from the D-instanton reference is the block in which the four non-compact fields X^μ, ψ^μ have mixed Dirichlet-Neumann boundary conditions. The instanton is Dirichlet in $\mathbb{R}^{1,3}$ and the D7-brane is Neumann in $\mathbb{R}^{1,3}$. The internal fractional CFT data are not replaced by a six-dimensional DN free field block; they enter through signed coefficients of extended $\mathcal{N} = 2$ characters.

In the reference convention, the contribution of a massive representation (h, Q) is

$$Z_{A,DN}^{(h,Q)} = \frac{(-1)^Q}{2} q^{h-(1+Q)/4} \vartheta_{1-Q,0}(2\tau). \quad (41)$$

After summing over representations, the annulus has the form

$$Z_{A,DN}(\tau) = \sum_{Q=0}^1 \frac{(-1)^Q}{2} \vartheta_{1-Q,0}(2\tau) \sum_{h \geq |Q|/2} n_{h,Q} q^{h-(1+Q)/4}. \quad (42)$$

Here $n_{h,Q}$ is the signed coefficient of representation (h, Q) in the character expansion of the open-string spectrum between the fractional instanton and the tadpole-cancelling space-filling branes, in the same orientation convention as (41). For a genuine massive representation this is an ordinary multiplicity. For massless representations it can be signed; for example the vacuum representation contributes +1 to $(h, Q) = (0, 0)$ and -1 to $(h, Q) = (1/2, 1)$ in the massive-character basis used by the reference.

For a single fractional instanton $\tilde{I}_a$ and D7 multiplicities N_m , this means

$$n_{h,Q} = \sum_{m=0}^2 N_m n_{h,Q}^{(a,m)}, \quad (43)$$

where $n_{h,Q}^{(a,m)}$ is computed from the eigenspace $\mathcal{H}_{am}^{IS}$ in (13). This is linear in the D7 multiplicities because one boundary is the single $O(1)$ instanton. If there are k_a identical instantons of label a , the right hand side is multiplied by k_a before the multi-instanton zero-mode analysis.

Equivalently, let $\mathcal{R}_{h,Q}^{IS}$ be the multiplicity space of the representation (h, Q) in the parent internal open-string Hilbert space $\mathcal{H}_{IS}$, with signed character decomposition understood as above. Then the fractional projection gives the exact coefficient

$$n_{h,Q}^{(a,m)} = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-(a-m)\ell} \text{Tr}_{\mathcal{R}_{h,Q}^{IS}}(g^\ell). \quad (44)$$

This formula is the corrected replacement for a double orbifold sum in the open channel. It projects the parent internal spectrum onto the eigenspace selected by the two fractional labels. It is written for the ordered sector $I \rightarrow S$. The reverse orientation is the ordered sector $S \rightarrow I$ and has

$$n_{h,Q}^{(m,a);SI} = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-(m-a)\ell} \text{Tr}_{\mathcal{R}_{h,Q}^{SI}}(g^\ell). \quad (45)$$

The APS block (41) is normalized so that the two open-string orientations are already paired when the two ordered sectors are related by charge conjugation. Thus the reverse orientation is not a second D7 boundary label and should not be represented by a second sum over D7 multiplicities.

8.3 Twined Character Data and $\mathbb{Z}_3$ Charge Sectors

The annulus data can be organized as a finite Fourier problem for each extended $\mathcal{N} = 2$ representation. For a massive representation (h, Q) in the parent ordered $I \rightarrow S$ internal Hilbert space, define the three twined traces

$$T_{h,Q}^{IS}(\ell) = \text{Tr}_{\mathcal{R}_{h,Q}^{IS}}(g^\ell), \quad \ell = 0, 1, 2, \quad (46)$$

where $\mathcal{R}_{h,Q}^{IS}$ is the multiplicity space used in (44). The corresponding $\mathbb{Z}_3$ -charge-sector multiplicities are

$$d_{h,Q;r}^{IS} = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-r\ell} T_{h,Q}^{IS}(\ell), \quad r \in \mathbb{Z}/3\mathbb{Z}. \quad (47)$$

Equivalently,

$$T_{h,Q}^{IS}(\ell) = \sum_{r=0}^2 d_{h,Q;r}^{IS} \omega^{r\ell}. \quad (48)$$

Here r is a worldsheet/orbifold eigenvalue label, not a transverse or spinor index. The fractional boundary labels select one of these three charge sectors:

$$n_{h,Q}^{(a,m)} = d_{h,Q;a-m}^{IS}, \quad a - m \text{ read in } \mathbb{Z}/3\mathbb{Z}. \quad (49)$$

Equation (49) is just (44) rewritten after the finite Fourier transform (47). Thus the missing massive annulus data are precisely the integers $d_{h,Q;r}^{IS}$, representation by representation.

For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$, the total DN coefficient for a single instanton label a simplifies to the untwined parent multiplicity:

$$n_{h,Q}^{(a)} = \sum_{m=0}^2 N_m n_{h,Q}^{(a,m)} = 8 \sum_{r=0}^2 d_{h,Q;r}^{IS} = 8 T_{h,Q}^{IS}(0). \quad (50)$$

This simplification applies to the non-zero-mode determinant once the zero-mode rows have been handled separately. It does not mean that the charged zero modes vanish; it only says that equal fractional D7 multiplicities remove the phase dependence from the total annulus coefficient.

The low-lying data already fixed in this note are the massless row analogues of (46):

$$T_{\text{vac}}^{IS}(\ell) = 1, \quad T_+^{IS}(\ell) = 3\omega^\ell, \quad T_-^{IS}(\ell) = 3\omega^{-\ell}. \quad (51)$$

Their charge-sector multiplicities are

$$d_{\text{vac};r}^{IS} = \delta_{r,0}, \quad d_{+;r}^{IS} = 3\delta_{r,1}, \quad d_{-;r}^{IS} = 3\delta_{r,-1}. \quad (52)$$

Substituting (52) into (49) reproduces the local zero-mode vector (134). The parent oscillator generating function in the next subsection fills the same charge-sector table for the massive representations that remain in Z_A^{det} , and the result resums to the A_2 -theta formula in (212).

8.4 Parent Oscillator Generating Function for the Twined Traces

The Fourier transform above reduces the annulus data to the parent twined traces $T_{h,Q}^{IS}(\ell)$. This subsection records the raw free-field oscillator trace from which those traces are extracted in the standard parent-boundary convention.

Assumption 4 (Parent internal oscillator convention). *For the involution-I ED3-D7 system, the fractional ED3 and the local D7 stack are taken to have the same internal parent boundary condition. Thus the internal ED3-D7 open string has no additional internal DN directions: the only DN directions are the four non-compact spacetime directions already accounted for in (41). In this boundary condition the parent internal oscillator algebra is generated by three complex bosons Z^i and three complex fermions ψ^i , $i = 1, 2, 3$, with the orbifold action in (5) and (133). If a different D7 boundary condition is specified, the generating function below must be replaced before computing K_0 .*

Let x be a formal variable that records the $\mathbb{Z}_3$ eigenvalue, let u record the internal $U(1)$ charge, let s record worldsheet fermion parity, and let q record oscillator energy above the NS open-string vacuum. The parent NS oscillators have the following weights:

oscillator	range	oscillator energy	x -weight	u -weight
α_{-n}^i	$n \geq 1$	n	x	1
$\bar{\alpha}_{-n}^{\bar{i}}$	$n \geq 1$	n	x^{-1}	1
ψ_{-r}^i	$r \in \mathbb{Z}_{\geq 0} + \frac{1}{2}$	r	x	u
$\bar{\psi}_{-r}^{\bar{i}}$	$r \in \mathbb{Z}_{\geq 0} + \frac{1}{2}$	r	x^{-1}	u^{-1}

The index i or $\bar{i}$ runs over 1, 2, 3. The variables x, u, s are bookkeeping variables only: x is reduced to $x = \omega^\ell$ when a twined trace is evaluated, u is used to decompose into internal $U(1)$ -charge sectors, and $s = -1$ inserts $(-1)^f$ in the NS oscillator trace.

The corresponding raw NS generating function is

$$\mathcal{P}_{\text{NS}}(x, u, s; q) = \prod_{n=1}^{\infty} \frac{1}{(1 - xq^n)^3 (1 - x^{-1}q^n)^3} \prod_{r \in \mathbb{Z}_{\geq 0} + \frac{1}{2}} (1 + suxq^r)^3 (1 + su^{-1}x^{-1}q^r)^3. \quad (53)$$

This is the parent oscillator trace before the Chan-Paton fractional projection, the GSO projection, the orientifold projection, and the subtraction of charged zero-mode representatives. The physical annulus coefficient is obtained only after these four operations are applied.

If $\epsilon_{\text{NS}} = +1$ keeps NS states with even worldsheet fermion parity and $\epsilon_{\text{NS}} = -1$ keeps NS states with odd worldsheet fermion parity, the NS GSO-projected raw trace in this convention is

$$\mathcal{P}_{\text{NS}}^{(\epsilon_{\text{NS}})}(x, u; q) = \frac{1}{2} [\mathcal{P}_{\text{NS}}(x, u, +1; q) + \epsilon_{\text{NS}} \mathcal{P}_{\text{NS}}(x, u, -1; q)]. \quad (54)$$

The sign ϵ_{NS} is not determined by (54); it must agree with the open-string GSO projection used in the BCFT zero-mode projection. With this GSO projection fixed, the raw NS multiplicity at oscillator energy E , internal charge J , and $\mathbb{Z}_3$ charge r is

$$p_{\text{NS}}^{(\epsilon_{\text{NS}})}(E, J; r) = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-r\ell} [q^E u^J] \mathcal{P}_{\text{NS}}^{(\epsilon_{\text{NS}})}(\omega^\ell, u; q), \quad (55)$$

where $[q^E u^J]$ means “coefficient of $q^E u^J$.” This formula is a finite $\mathbb{Z}_3$ Fourier projection of the parent oscillator trace. It should not be confused with $d_{h,Q;r}^{IS}$ in (47): p_{NS} counts raw oscillator states, while $d_{h,Q;r}^{IS}$ counts multiplicity spaces of extended $\mathcal{N} = 2$ representations after the character decomposition.

The character extraction is triangular in the q -expansion. In the stated normalization of the internal NS characters, write

$$\mathcal{P}_{\text{NS}}^{(\epsilon_{\text{NS}})}(\omega^\ell, u; q) = \sum_{\lambda \in \{\text{vac}, +, -\}} A_\lambda^{IS}(\ell) \chi_{\lambda, \text{NS}}(u; q) + \sum_{(h, Q)} T_{h, Q}^{IS}(\ell) \chi_{h, Q, \text{NS}}(u; q). \quad (56)$$

Here $\chi_{\lambda, \text{NS}}$ is the NS character of the massless internal representation $\lambda \in \{\text{vac}, +, -\}$, $\chi_{h, Q, \text{NS}}$ is the NS character of the massive representation labelled by (h, Q) , and $A_\lambda^{IS}(\ell)$ are the twined coefficients of the three massless rows. The massive coefficients in (56) are precisely the traces $T_{h, Q}^{IS}(\ell)$ used in (46). The extraction is triangular because each character has a definite lowest power of q ; after subtracting all characters that begin below a given power, the coefficient at the next lowest power fixes the next multiplicity.

As a check on the normalization, the first terms of the raw NS generating function are

$$\mathcal{P}_{\text{NS}}(x, u, s; q) = 1 + 3sux q^{1/2} + 3su^{-1}x^{-1}q^{1/2} + \left[3(x + x^{-1}) + 9 + 3u^2x^2 + 3u^{-2}x^{-2} \right] q + O(q^{3/2}). \quad (57)$$

Setting $x = \omega^\ell$ reproduces the low-lying twined row traces in (51). Thus the oscillator trace and the local row analysis use the same $\mathbb{Z}_3$ -charge convention. The q -order terms are displayed to make the next step concrete: they are raw oscillator states with $(J, r) = (0, 1), (0, -1), (0, 0), (2, -1)$, and $(-2, 1)$, respectively, where J is the internal $U(1)$ charge and r is the $\mathbb{Z}_3$ charge. They are not representation multiplicities: descendants of the massless characters in (56) must first be subtracted.

8.5 Ramond Ground States and the Spin Lift

The Ramond sector requires one more piece of local BCFT data: the lift of the orbifold generator g to spin fields. This lift is easy to miss because g has order three on vectors but its naive spin representative has order six.

Bosonize the three complex internal fermions by

$$\psi^i = e^{iH_i}, \quad \bar{\psi}^{\bar{i}} = e^{-iH_i}, \quad i = 1, 2, 3. \quad (58)$$

The Ramond ground states are spin fields

$$S_\varepsilon = \exp \left[\frac{i}{2} \sum_{i=1}^3 \varepsilon_i H_i \right], \quad \varepsilon_i = \pm 1. \quad (59)$$

Define

$$J(\varepsilon) = \frac{1}{2} \sum_{i=1}^3 \varepsilon_i, \quad n_-(\varepsilon) = \#\{i : \varepsilon_i = -1\}. \quad (60)$$

Here $J(\varepsilon) = 3/2 - n_-(\varepsilon)$ is the internal $U(1)$ charge of the spin field. The vector action $\psi^i \mapsto \omega \psi^i$ is represented on bosons by $H_i \mapsto H_i + 2\pi/3$. The naive spin-field phase is therefore $\exp(2\pi i J(\varepsilon)/3)$, whose cube is -1 on every spin field. To obtain an order-three action on the Ramond ground states, multiply this naive lift by a phase $-\omega^\rho$, with

$$\rho \in \mathbb{Z}/3\mathbb{Z}. \quad (61)$$

The resulting spin lift is

$$g_R^{(\rho)} S_\varepsilon = \omega^{r_\rho(\varepsilon)} S_\varepsilon, \quad r_\rho(\varepsilon) = \rho - n_-(\varepsilon) \pmod{3}. \quad (62)$$

The parameter ρ is not a transverse index. It is a discrete spin-lift datum to be fixed together with the closed-string GSO projection and the orientifold lift. The common Calabi-Yau lift is $\rho = 0$, for which the two $SU(3)$ -singlet spin fields with $n_- = 0$ and $n_- = 3$ have $\mathbb{Z}_3$ charge zero. The note does not assume this choice unless it is stated explicitly.

The Ramond ground-state charge polynomial in this convention is

$$\mathcal{G}_R^{(\rho)}(x, u) = x^\rho (u^{3/2} + u^{-3/2}) + 3x^{\rho-1} u^{1/2} + 3x^{\rho-2} u^{-1/2}, \quad (63)$$

where powers of x are read modulo 3 after setting $x = \omega^\ell$. Equivalently, the Ramond ground-state charge-sector multiplicities are

$n_-(\varepsilon)$	$J(\varepsilon)$	multiplicity	$\mathbb{Z}_3$ charge r
0	3/2	1	ρ
1	1/2	3	$\rho - 1$
2	-1/2	3	$\rho - 2$
3	-3/2	1	ρ

(64)

all charges being in $\mathbb{Z}/3\mathbb{Z}$. For a fractional ED3 of label a and a D7 label m , the orbifold projection keeps the row $r = a - m$. Thus the raw Ramond ground states in $\mathcal{H}_{am}^{IS}$ are selected by

$$m = a - r_\rho(\varepsilon) \pmod{3}. \quad (65)$$

After fixing the R-sector GSO parity χ_R , it is useful to record only the spin fields that survive that parity. Define

$$s_{\rho, \chi_R}(r) = \#\{\varepsilon \mid n_-(\varepsilon) \equiv \chi_R \pmod{2}, \quad r_\rho(\varepsilon) = r\}, \quad r \in \mathbb{Z}/3\mathbb{Z}. \quad (66)$$

Here $s_{\rho, \chi_R}(r)$ is a multiplicity of internal Ramond spin fields. The corresponding fermionic $L_0 = 0$ vertex dimension is obtained only after imposing the orientifold action and the external vertex factors. The charge table (64) gives

	$s_{\rho, \chi_R}(\rho)$	$s_{\rho, \chi_R}(\rho - 1)$	$s_{\rho, \chi_R}(\rho - 2)$
$\chi_R = 0$	1	0	3
$\chi_R = 1$	1	3	0

(67)

with the three charge labels read modulo 3. Equivalently, for the common Calabi-Yau spin lift $\rho = 0$,

$$\begin{array}{c|ccc} & r=0 & r=1 & r=2 \\ \hline \chi_R = 0 & 1 & 3 & 0 \\ \chi_R = 1 & 1 & 0 & 3 \end{array} \quad (68)$$

This last display is only the $\rho = 0$ specialization; the formulas above keep ρ explicit.

The same table gives a useful orientifold compatibility test. Suppose the R-sector orientifold maps the source branch (ρ, χ_R) in the ordered $I \rightarrow S$ sector to a target branch $(\rho_\Theta, \chi_R^\Theta)$ in the ordered $S \rightarrow I$ sector and conjugates the fractional charge $r \mapsto -r$. A finite invertible spin-field matrix can exist only if

$$s_{\rho, \chi_R}(r) = s_{\rho_\Theta, \chi_R^\Theta}(-r), \quad r = 0, 1, 2. \quad (69)$$

Here $-r$ is read in $\mathbb{Z}/3\mathbb{Z}$. Comparing the two rows of (67) gives

$$\rho_\Theta = -\rho \pmod{3}, \quad \chi_R^\Theta = \chi_R + 1 \pmod{2}. \quad (70)$$

Thus a charge-conjugating orientifold that acts with the same spin-lift label on source and target forces $\rho = 0$, and it exchanges the two internal parity branches $\chi_R = 0$ and $\chi_R = 1$. This is a statement about the internal Ramond spin fields only. If the full open-string GSO projection packages the parity exchange with external or ghost factors, the result must still satisfy (69).

Equivalently, before the external DN spin fields, GSO projection, and orientifold projection are imposed, the ordered $I \rightarrow S$ Ramond ground-state shadows are

$$W_{R,J}^{IS}(a; \rho) = \bigoplus_{\varepsilon: J(\varepsilon)=J} \mathbb{C} S_\varepsilon \otimes V_{a-r_\rho(\varepsilon)}. \quad (71)$$

Here V_m is the D7 Chan-Paton space of fractional label m , with labels read modulo 3. Their dimensions are

$$\begin{array}{c|c} J & \dim W_{R,J}^{IS}(a; \rho) \\ \hline 3/2 & N_{a-\rho} \\ 1/2 & 3N_{a-\rho+1} \\ -1/2 & 3N_{a-\rho+2} \\ -3/2 & N_{a-\rho} \end{array} \quad (72)$$

This is the Ramond analogue of the local vector (134). It is a ground-state multiplicity before the total open-string GSO projection, not a fermionic zero-mode count. The two entries with $J = \pm 3/2$ are two internal spin-field rows; depending on the external DN spin field and picture, one linear combination or chirality may be kept.

The nonzero Ramond oscillators have integer modes. Their oscillator generating function, before the GSO and orientifold projections, is

$$\mathcal{P}_R^{(\rho)}(x, u, s; q) = \mathcal{G}_R^{(\rho)}(x, u) \prod_{n=1}^{\infty} \frac{(1 + suxq^n)^3 (1 + su^{-1}x^{-1}q^n)^3}{(1 - xq^n)^3 (1 - x^{-1}q^n)^3}. \quad (73)$$

Here s records the parity of nonzero fermion oscillator excitations. The action of the full open-string GSO operator on the Ramond ground states is not determined by (73) alone, because it also involves the non-compact DN spin fields and the specified picture. Consequently the R-sector charged zero-mode spaces $\mathcal{B}_{\lambda,R}^{\text{proj}}(a)$ must still be evaluated as vertex spaces, not inferred only from $\mathcal{G}_R^{(\rho)}$.

The first nonzero-oscillator terms are

$$\mathcal{P}_R^{(\rho)}(x, u, s; q) = \mathcal{G}_R^{(\rho)}(x, u) \left[1 + 3 \left(x + x^{-1} + sux + su^{-1}x^{-1} \right) q + O(q^2) \right]. \quad (74)$$

The terms $3xq$ and $3x^{-1}q$ come from the bosonic oscillators α_{-1}^i and $\bar{\alpha}_{-1}^{\bar{i}}$, while the terms with s come from the integer-moded R-sector fermion oscillators ψ_{-1}^i and $\bar{\psi}_{-1}^{\bar{i}}$. This is a raw oscillator expansion before the GSO projection and before the Ramond vertex-space constraints.

For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$, the dependence on a and ρ drops out of the total raw Ramond ground-state count before the GSO and orientifold projections:

$$\sum_{m=0}^2 N_m \sum_{\varepsilon: r_\rho(\varepsilon)=a-m} 1 = 8 \sum_{\varepsilon} 1 = 64. \quad (75)$$

This number is only a raw internal spin-field count. It is not f_{ch} , not an annulus coefficient, and not a determinant contribution until the external DN spin fields, GSO projection, orientifold projection, and BRST vertex representatives have been included.

8.6 External DN Boundary-changing Factors

The internal generating functions above must be tensored with the universal four-dimensional DN boundary changer. This subsection records that factor at the level needed for the zero-mode calculation. It is local vertex data, not a global annulus formula.

Let Δ_{IS} denote the bosonic boundary-changing operator that changes the four non-compact coordinates X^μ , $\mu = 0, 1, 2, 3$, from Dirichlet on the instanton boundary to Neumann on the D7 boundary. Let Δ_{SI} denote the reverse-orientation operator. For four real DN bosons,

$$h(\Delta_{IS}) = h(\Delta_{SI}) = \frac{4}{16} = \frac{1}{4}. \quad (76)$$

Here h is the holomorphic boundary conformal weight. The operator Δ_{IS} has no $\mathbb{Z}_3$ charge because the orbifold acts only on the internal coordinates z_i .

In the NS sector, the four DN fermions have zero modes. They generate a Clifford module for the rotation group of the four non-compact DN directions. We denote the two states of one chirality by

$$S_{\text{DN}}^{\mathbf{s}}, \quad \mathbf{s} = 1, 2, \quad h(S_{\text{DN}}^{\mathbf{s}}) = \frac{1}{4}. \quad (77)$$

The index $\mathbf{s}$ is an external DN spinor index, not an orbifold label and not an internal $SU(3)$ index. Which chirality is represented by $S_{\text{DN}}^{\mathbf{s}}$ is fixed by the GSO branch; the opposite chirality is removed by $P_{\text{GSO,NS}}$.

Let ϕ be the bosonized superghost, normalized so that

$$h(e^{-\phi}) = \frac{1}{2}, \quad h(e^{-\phi/2}) = \frac{3}{8}. \quad (78)$$

An ordered $I \rightarrow S$ NS zero-mode vertex in the (-1) -picture has the form

$$V_u^{(-1)} = u_{\mathbf{s},A} \Delta_{IS} S_{\text{DN}}^{\mathbf{s}} \mathcal{O}_A^{\text{NS}} e^{-\phi}, \quad (79)$$

where $u_{\mathbf{s},A}$ is the coefficient of the zero-mode vertex and $\mathcal{O}_A^{\text{NS}}$ is an internal boundary operator, including its Chan-Paton label. The condition that (79) have total boundary weight one is

$$h(\mathcal{O}_A^{\text{NS}}) = 0. \quad (80)$$

Thus, in the standard four-DN vertex construction, bosonic charged zero modes come from internal NS operators of weight zero. Internal NS operators of higher weight can still enter the massive annulus character expansion, but they are not $L_0 = 0$ bosonic zero-mode vertices of the form (79).

In the R sector, the DN fermions are half-integer moded and do not create an external spinor zero-mode degeneracy. An ordered $I \rightarrow S$ R zero-mode vertex in the $(-1/2)$ -picture has the form

$$V_\eta^{(-1/2)} = \eta_A \Delta_{IS} \mathcal{S}_A^R e^{-\phi/2}, \quad (81)$$

where η_A is the Grassmann coefficient of the zero-mode vertex and $\mathcal{S}_A^R$ is an internal Ramond boundary operator, including its Chan-Paton label. The weight-one condition is

$$h(\mathcal{S}_A^R) = \frac{3}{8}. \quad (82)$$

The internal spin fields S_ε in (59) have this weight. Therefore the charge table (64) is exactly the internal Ramond ground-state input to the R-sector charged zero-mode calculation.

For later reference, the vertex-level zero-mode test can be split into a fixed kinematic part and a projection part fixed by the GSO choice and orientifold lift:

sector	external factor	internal weight required	projection still required
NS	$\Delta_{IS} S_{\text{DN}}^s e^{-\phi}$	$h_{\text{int}} = 0$	$P_{\text{GSO,NS}}, \Theta_{\text{NS}}$
R	$\Delta_{IS} e^{-\phi/2}$	$h_{\text{int}} = 3/8$	$P_{\text{GSO,R}}, \Theta_R$

(83)

Here Θ_{NS} and Θ_R are the orientifold maps on the full vertex operators, including the exchange $\Delta_{IS} \leftrightarrow \Delta_{SI}$, the action on S_{DN}^s in the NS sector, the internal action, and the Chan-Paton matrices. These signs and matrices are not determined by the conformal weights in this subsection.

8.7 Zero-mode Spaces After the DN Weight Test

The vertex templates above refine the earlier row-level bookkeeping. Under Assumption 4, the only internal NS operator of weight zero is the identity row. Therefore the ordered $I \rightarrow S$ NS charged zero-mode shadow is

$$\mathcal{Z}_{\text{NS}}^{IS}(a) = \mathbb{C}_{\text{DN,NS}}^2 \otimes V_a, \quad \dim \mathcal{Z}_{\text{NS}}^{IS}(a) = 2N_a. \quad (84)$$

Here $\mathbb{C}_{\text{DN,NS}}^2$ is spanned by the two external DN spin fields S_{DN}^s that survive the fixed NS GSO chirality, and V_a is the D7 Chan-Paton space with the same fractional label as the instanton. The adjacent rows generated by ψ^i and $\bar{\psi}^{\bar{i}}$ have internal weight $1/2$, so they are not NS $L_0 = 0$ zero modes in the standard four-DN vertex construction. They belong in the non-zero-mode determinant unless another vertex construction or boundary condition is specified.

For the R sector, introduce a parity label

$$\chi_R \in \mathbb{Z}/2\mathbb{Z} \quad (85)$$

for the internal Ramond spin fields kept by the open-string GSO projection: in this notation $n_-(\varepsilon) \equiv \chi_R \pmod{2}$ is kept. This is only a label for the two possible GSO branches; the physical branch must be fixed by the preserved four-dimensional supersymmetry and the orientifold lift. Before the orientifold projection, the ordered $I \rightarrow S$ R charged zero-mode shadow is

$$\mathcal{Z}_R^{IS}(a; \rho, \chi_R) = \bigoplus_{\varepsilon: n_-(\varepsilon) \equiv \chi_R \pmod{2}} \mathbb{C} S_\varepsilon \otimes V_{a-r_\rho(\varepsilon)}. \quad (86)$$

Its dimension is

$$\dim \mathcal{Z}_R^{IS}(a; \rho, \chi_R) = \begin{cases} N_{a-\rho} + 3N_{a-\rho+2}, & \chi_R = 0, \\ 3N_{a-\rho+1} + N_{a-\rho}, & \chi_R = 1, \end{cases} \quad (87)$$

with all fractional labels read in $\mathbb{Z}/3\mathbb{Z}$.

The reverse ordered sector $S \rightarrow I$ has the charge-conjugate Chan-Paton representation. If the orientifold acts by an off-diagonal orientation-pair involution and commutes with the GSO projection, the invariant subspace in $\mathcal{Z}^{IS} \oplus \mathcal{Z}^{SI}$ has the same dimension as the ordered space $\mathcal{Z}^{IS}$. Thus, in that ordinary projection convention,

$$\dim \mathcal{Z}_{\text{NS}}^{\text{proj}}(a) = 2N_a, \quad \dim \mathcal{Z}_R^{\text{proj}}(a; \rho, \chi_R) = \dim \mathcal{Z}_R^{IS}(a; \rho, \chi_R). \quad (88)$$

Equation (88) is a dimension statement about the ordinary orientation-pair projection. It does not fix the detailed orientifold signs needed for non-vacuum Mobius coefficients.

For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$, the same formula gives

$$\dim \mathcal{Z}_{\text{NS}}^{\text{proj}}(a) = 16, \quad \dim \mathcal{Z}_R^{\text{proj}}(a; \rho, \chi_R) = 32 \quad (89)$$

under the ordinary orientation-pair projection, independently of a , ρ , and χ_R . In particular, the standalone K_0 formula is already obstructed by the NS charged bosonic zero modes whenever $N_a > 0$ in this convention.

8.8 Dimension Counts Versus Determinant Weights

The dimensions in (88) are dimensions of $L_0 = 0$ vertex spaces. They are not, by themselves, the constants that must be subtracted from the annulus integrand. The annulus uses the gauge-fixed supertrace with $b_0 c_0$ and the same ghost convention as (148). Therefore the determinant subtraction is a weighted trace, not an ordinary boson-minus-fermion dimension.

For each projected zero-mode sector define its gauge-fixed determinant weight by

$$W_{\lambda, \nu}^0(a) = \text{STr}_{\mathcal{B}_{\lambda, \nu}^{\text{proj}}(a)}^{\text{gf}}(b_0 c_0), \quad \lambda \in \{\text{vac}, \text{adj}\}, \quad \nu \in \{\text{NS}, \text{R}\}. \quad (90)$$

Here the GSO and orientifold projections have already been imposed in the definition of $\mathcal{B}_{\lambda, \nu}^{\text{proj}}(a)$. In the covering-space convention of (148), the same number is computed by inserting the orientifold average and the projector P_{GSO} before taking the trace.

The total charged zero-mode constant to remove from the large- t annulus is

$$Z_{A, \text{ch}}^0(a) = \sum_{\lambda \in \{\text{vac}, \text{adj}\}} \left[W_{\lambda, \text{NS}}^0(a) + W_{\lambda, \text{R}}^0(a) \right]. \quad (91)$$

Equation (91) is the weighted version of the dimension count in (88). It is the $L_0 = 0$ constant to remove from the annulus determinant.

The identity block gives a useful normalization check. From (158),

$$\lim_{t \rightarrow \infty} Z_{A, \text{DN}}^{(a), \text{id}}(t) = \frac{3}{2} N_a. \quad (92)$$

This is a gauge-fixed determinant weight of the internal identity representation, not the ordinary dimension $2N_a$ of the NS zero-mode space in (84). The mismatch is expected: the APS trace includes ghost and picture data. Consequently, once the explicit BRST representatives are chosen, their weights must be checked against (92); one must not replace $Z_{A, \text{ch}}^0$ by $b_{\text{ch}} - f_{\text{ch}}$.

8.9 Finite Matrix Form of the Zero-mode Projection

The previous subsections reduce the charged zero-mode question to a finite problem. This subsection writes that finite problem in a basis. The purpose is to make clear which data are already fixed and which data still have to be supplied by the explicit orientifold BCFT.

Assume in this subsection that the orientifold lift has a one-node invariant instanton label, so $s = 2a$ in (18). Let

$$E_S = \bigoplus_{m=0}^2 V_m, \quad \Gamma_S : E_S \longrightarrow E_{\Omega_\sigma(S)}^\vee \quad (93)$$

be the total D7 Chan-Paton space and its orientifold Chan-Paton map. For $r \in \mathbb{Z}/3\mathbb{Z}$, let

$$\Gamma_{S,r} : V_{a-r} \longrightarrow V_{a+r}^\vee \quad (94)$$

be the restriction of Γ_S to the pair of image fractional labels $a - r \leftrightarrow a + r$. The target is a dual space because worldsheet parity reverses the orientation of the open string. In particular, $r = 0$ uses $\Gamma_{S,0} : V_a \rightarrow V_a^\vee$.

Let

$$E_{\text{DN,NS}} = \text{span}\{S_{\text{DN}}^{\mathbf{s}} : \mathbf{s} = 1, 2\} \quad (95)$$

be the external DN spinor space that survived the fixed NS GSO chirality in (77). The most general ordinary off-diagonal orientifold action on the standard NS zero-mode representatives has the form

$$U_{\text{NS}} = \xi_{\text{NS}} C_{\text{DN,NS}} \otimes \Gamma_{S,0} : E_{\text{DN,NS}} \otimes V_a \longrightarrow E_{\text{DN,NS}}^{SI} \otimes V_a^\vee. \quad (96)$$

Here $E_{\text{DN,NS}}^{SI}$ is the corresponding external spinor space in the reverse ordered sector, $C_{\text{DN,NS}}$ is the finite matrix by which Ω_σ acts on the two DN fermion zero-mode spin fields, and ξ_{NS} is the product of the phases from the exchange $\Delta_{IS} \leftrightarrow \Delta_{SI}$, the superghost representative, and any fixed worldsheet-parity convention. Equation (96) is not an assumption that these phases are known; it is the finite matrix whose entries must be computed or specified.

For the R sector it is cleaner to group spin fields by their $\mathbb{Z}_3$ charge. Define

$$\mathcal{S}_R^{(\rho, \chi_R)}(r) = \text{span}\{S_\varepsilon \mid n_-(\varepsilon) \equiv \chi_R \pmod{2}, \quad r_\rho(\varepsilon) = r\}, \quad r \in \mathbb{Z}/3\mathbb{Z}, \quad (97)$$

where $r_\rho(\varepsilon)$ was defined in (62). Then (86) may be rewritten as

$$\mathcal{Z}_R^{IS}(a; \rho, \chi_R) = \bigoplus_{r \in \mathbb{Z}/3\mathbb{Z}} \mathcal{S}_R^{(\rho, \chi_R)}(r) \otimes V_{a-r}. \quad (98)$$

The orientifold row orbits for the R-sector charged zero modes are the charge-zero row and the adjacent charge pair:

$$\mathcal{Z}_{R,\text{vac}}^{IS}(a; \rho, \chi_R) = \mathcal{S}_R^{(\rho, \chi_R)}(0) \otimes V_a, \quad \mathcal{Z}_{R,\text{adj}}^{IS}(a; \rho, \chi_R) = \left(\mathcal{S}_R^{(\rho, \chi_R)}(1) \otimes V_{a-1}\right) \oplus \left(\mathcal{S}_R^{(\rho, \chi_R)}(2) \otimes V_{a+1}\right). \quad (99)$$

The labels vac and adj in (99) refer to the fractional charge row $r = 0$ and the orientifold-paired adjacent rows $r = \pm 1$. They are row orbit labels, not transverse or spinor indices. Their ordered-sector dimensions are

$$\dim \mathcal{Z}_{R,\text{vac}}^{IS} = s_{\rho, \chi_R}(0)N_a, \quad \dim \mathcal{Z}_{R,\text{adj}}^{IS} = s_{\rho, \chi_R}(1)N_{a-1} + s_{\rho, \chi_R}(2)N_{a+1}. \quad (100)$$

For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$, the total R dimension after the ordinary orientation-pair projection is always

$$8 \sum_{r=0}^2 s_{\rho, \chi_R}(r) = 32, \quad (101)$$

but the split between the $r = 0$ row and the adjacent pair depends on ρ and χ_R . For $\rho = 0$, (68) gives

$$\dim \mathcal{Z}_{R, \text{vac}}^{\text{proj}} = 8, \quad \dim \mathcal{Z}_{R, \text{adj}}^{\text{proj}} = 24, \quad (102)$$

for either GSO parity branch $\chi_R = 0$ or $\chi_R = 1$, with the adjacent label $r = 1$ or $r = 2$ determined by the branch.

Let $(\rho_\Theta, \chi_R^\Theta)$ be the Ramond branch reached by the orientifold in the reverse ordered sector. In the ordinary charge-conjugating convention, the compatibility condition (69) gives the branch map (70). We then write the finite spin-field matrix as

$$\mathbf{M}_R(r) : \mathcal{S}_R^{(\rho, \chi_R)}(r) \longrightarrow \mathcal{S}_{R, SI}^{(\rho_\Theta, \chi_R^\Theta)}(-r), \quad (103)$$

where the subscript SI reminds the reader that the target is the reverse ordered sector. The corresponding orientifold map on the ordered R zero-mode representatives is

$$\mathbf{U}_R|_{\mathcal{S}_R^{(\rho, \chi_R)}(r) \otimes V_{a-r}} = \xi_R \mathbf{M}_R(r) \otimes \Gamma_{S, r}. \quad (104)$$

The phase ξ_R includes the action on Δ_{IS} , the $(-1/2)$ -picture superghost representative, and the Ramond-sector worldsheet-parity convention. If the explicit orientifold does not act by the branch map (70) and charge conjugation $r \mapsto -r$, then (103) must be replaced by the actual branch and charge map before any projection or determinant weight is evaluated.

For the ordinary Calabi-Yau spin lift $\rho = 0$, and for the standard charge-conjugating action on internal spin labels, the preceding abstract branch map can be written in an explicit spin-field basis. Let

$$S_+ = S_{(+, +, +)}, \quad S_- = S_{(-, -, -)}, \quad (105)$$

and, for $i = 1, 2, 3$, let $S_i^{(1)}$ be the spin field with the single minus sign in the i -th entry and let $S_i^{(2)}$ be the spin field with the single plus sign in the i -th entry:

$$S_i^{(1)} = S_{\varepsilon_i^{(1)}}, \quad \varepsilon_{i,j}^{(1)} = \begin{cases} -1, & j = i, \\ +1, & j \neq i, \end{cases} \quad S_i^{(2)} = S_{-\varepsilon_i^{(1)}}. \quad (106)$$

The superscripts (1) and (2) record $n_-(\varepsilon) = 1$ and $n_-(\varepsilon) = 2$, respectively; they are not powers. In this basis

$$\mathcal{S}_R^{(0,0)}(0) = \mathbb{C}S_+, \quad \mathcal{S}_R^{(0,0)}(1) = \text{span}\{S_i^{(2)}\}_{i=1}^3, \quad \mathcal{S}_R^{(0,0)}(2) = 0, \quad (107)$$

$$\mathcal{S}_R^{(0,1)}(0) = \mathbb{C}S_-, \quad \mathcal{S}_R^{(0,1)}(1) = 0, \quad \mathcal{S}_R^{(0,1)}(2) = \text{span}\{S_i^{(1)}\}_{i=1}^3. \quad (108)$$

Thus the two ordinary Ramond branches are paired as follows:

$$\begin{aligned} \chi_R = 0 : \quad & r = 0 : \mathbb{C}S_+ \longmapsto \chi_R^\Theta = 1, \quad r = 0 : \mathbb{C}S_-^{SI}, \\ & r = 1 : \text{span}\{S_i^{(2)}\} \longmapsto \chi_R^\Theta = 1, \quad r = 2 : \text{span}\{S_i^{(1), SI}\}, \\ \chi_R = 1 : \quad & r = 0 : \mathbb{C}S_- \longmapsto \chi_R^\Theta = 0, \quad r = 0 : \mathbb{C}S_+^{SI}, \\ & r = 2 : \text{span}\{S_i^{(1)}\} \longmapsto \chi_R^\Theta = 0, \quad r = 1 : \text{span}\{S_i^{(2), SI}\}. \end{aligned} \quad (109)$$

The entries in the table are spin-field spaces only. The corresponding Chan-Paton spaces are attached by (98): a source row of charge r carries V_{a-r} , and the ordinary orientifold sends it to the reverse ordered row of charge $-r$ with $V_{a+r}^\vee$ through $\Gamma_{S,r}$.

Consequently the finite Ramond matrices that remain to be computed in this ordinary $\rho = 0$ convention are

$$\mathbf{M}_R^{(0)}(0) : \mathbb{C}S_+ \longrightarrow \mathbb{C}S_-^{SI}, \quad \mathbf{M}_R^{(0)}(1) : \text{span}\{S_i^{(2)}\} \longrightarrow \text{span}\{S_i^{(1),SI}\}, \quad (110)$$

$$\mathbf{M}_R^{(1)}(0) : \mathbb{C}S_- \longrightarrow \mathbb{C}S_+^{SI}, \quad \mathbf{M}_R^{(1)}(2) : \text{span}\{S_i^{(1)}\} \longrightarrow \text{span}\{S_i^{(2),SI}\}. \quad (111)$$

The superscript on $\mathbf{M}_R^{(b)}$ records the source parity branch $b = \chi_R$. The one-dimensional maps in (110)–(111) are phases, and the three-dimensional maps are finite invertible matrices. Their entries depend on the orientifold action on spin fields, the cocycles in the bosonized fermion system, and the superghost normalization. The table fixes their source and target spaces; it does not fix these phases or matrices.

For either $\nu = \text{NS}$ or $\nu = R$, let $\mathbf{U}_\nu$ denote the appropriate ordered-sector map above. The covering orientation-pair action is

$$\Theta_\nu(z_{IS}, z_{SI}) = \left(\epsilon_\nu \mathbf{U}_\nu^{-1} z_{SI}, \epsilon_\nu \mathbf{U}_\nu z_{IS} \right), \quad \epsilon_\nu = \pm 1, \quad (112)$$

where z_{IS} is an ordered $I \rightarrow S$ zero-mode representative and z_{SI} is its reverse ordered partner. The sign ϵ_ν is the overall orientifold parity of the chosen vertex basis. The compatibility condition $\Theta_\nu^2 = 1$ is a finite matrix condition on ϵ_ν , $\mathbf{U}_\nu$, and the Chan-Paton maps.

When (112) is an off-diagonal involution, its invariant subspace is

$$\mathcal{Z}_\nu^{\text{proj}} = \left\{ (z, \epsilon_\nu \mathbf{U}_\nu z) \mid z \in \mathcal{Z}_\nu^{IS} \right\}. \quad (113)$$

Thus the finite matrices in (96) and (104) fix the orientifold-invariant zero-mode basis, but they do not by themselves remove a nonzero orientation-pair zero-mode space.

The determinant weights are computed in the same finite basis. Let $\mathbf{w}_\nu$ denote the gauge-fixed one-particle weight operator induced by $(-1)^F b_0 c_0$, including the R-sector normalization associated with the G_0 kinetic operator when $\nu = R$. In the orientifold-average convention of (148),

$$W_\nu^0(a) = \frac{1}{2} \text{Tr}_{\mathcal{Z}_\nu^{IS} \oplus \mathcal{Z}_\nu^{SI}} [\mathbf{w}_\nu P_{\text{GSO},\nu} (1 + \Theta_\nu)]. \quad (114)$$

For a purely off-diagonal Θ_ν , the trace of the Θ_ν term vanishes in the ordered-pair basis, and (114) is equal to the same weighted trace on one ordered sector. This explains why orientifold signs can be crucial for the surviving vertex basis while leaving the annulus zero-mode constant unchanged in the ordinary orientation-pair projection.

Under the standard four-DN vertex construction used in (84), the NS vertex-space part of the table is already fixed, and the APS identity-row determinant weight gives a row-level constraint:

$$\dim \mathcal{B}_{\text{vac,NS}}^{\text{proj}}(a) = 2N_a, \quad \mathcal{B}_{\text{adj,NS}}^{\text{proj}}(a) = 0, \quad W_{\text{vac,APS}}^0(a) = \frac{3}{2} N_a. \quad (115)$$

Here $W_{\text{vac,APS}}^0$ denotes the total gauge-fixed large- t constant of the internal identity representation in the APS spin-structure sum. It is exactly (92). It is not, by itself, a decomposition into NS-only and R-only determinant weights. The equality $\mathcal{B}_{\text{adj,NS}}^{\text{proj}} = 0$ says that the adjacent ψ^i and $\bar{\psi}^{\bar{i}}$ rows have internal weight $1/2$, so they are not NS $L_0 = 0$ representatives in the standard vertex (79).

The charged Ramond zero-mode determinant input is finite-dimensional. Write

$$W_R^0(a; \rho, \chi_R) = W_{\text{vac},R}^0(a; \rho, \chi_R) + W_{\text{adj},R}^0(a; \rho, \chi_R) \quad (116)$$

for the total Ramond charged zero-mode determinant weight. If $w_R(\varepsilon)$ denotes the gauge-fixed weight of the one-dimensional representative associated with S_ε , then the row split in (99) gives

$$W_{\text{vac},R}^0(a; \rho, \chi_R) = \sum_{\substack{\varepsilon: n_-(\varepsilon) \equiv \chi_R \pmod{2} \\ r_\rho(\varepsilon)=0}} w_R(\varepsilon) N_a, \quad (117)$$

$$W_{\text{adj},R}^0(a; \rho, \chi_R) = \sum_{\substack{\varepsilon: n_-(\varepsilon) \equiv \chi_R \pmod{2} \\ r_\rho(\varepsilon)=1}} w_R(\varepsilon) N_{a-1} + \sum_{\substack{\varepsilon: n_-(\varepsilon) \equiv \chi_R \pmod{2} \\ r_\rho(\varepsilon)=2}} w_R(\varepsilon) N_{a+1}. \quad (118)$$

Adding (117) and (118), the ordinary orientation-pair projection gives

$$W_R^0(a; \rho, \chi_R) = \sum_{\varepsilon: n_-(\varepsilon) \equiv \chi_R \pmod{2}} w_R(\varepsilon) N_{a-r_\rho(\varepsilon)}. \quad (119)$$

For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$, this becomes

$$W_R^0(a; \rho, \chi_R) = 8 \sum_{\varepsilon: n_-(\varepsilon) \equiv \chi_R \pmod{2}} w_R(\varepsilon). \quad (120)$$

In the ordinary $\rho = 0$ branch basis (107)–(108), the same finite sum can be written with four types of Ramond weights:

$$w_+ = w_R(S_+), \quad w_- = w_R(S_-), \quad w_i^{(1)} = w_R(S_i^{(1)}), \quad w_i^{(2)} = w_R(S_i^{(2)}). \quad (121)$$

Here $S_\pm$, $S_i^{(1)}$, and $S_i^{(2)}$ are the spin fields defined in (105)–(106); the index $i = 1, 2, 3$ is a label of the three complex internal fermions. For the even Ramond branch $\chi_R = 0$,

$$W_R^0(a; 0, 0) = w_+ N_a + \left(\sum_{i=1}^3 w_i^{(2)} \right) N_{a-1}, \quad (122)$$

whereas for the odd Ramond branch $\chi_R = 1$,

$$W_R^0(a; 0, 1) = w_- N_a + \left(\sum_{i=1}^3 w_i^{(1)} \right) N_{a+1}. \quad (123)$$

If the bootstrap-fixed orientifold action and cocycles preserve the permutation symmetry of the three complex internal fermions, then the three $w_i^{(1)}$ are equal to a common weight w_1 , and the three $w_i^{(2)}$ are equal to a common weight w_2 . Under that additional symmetry assumption, the symmetric tadpole solution gives

$$W_R^0(a; 0, 0) = 8(w_+ + 3w_2), \quad W_R^0(a; 0, 1) = 8(w_- + 3w_1). \quad (124)$$

Equations (119)–(124) define the finite Ramond data needed if one computes the full charged-mode measure. For the standalone K_0 question, the NS identity row already obstructs the ordinary local D7 model whenever $N_a > 0$. The Ramond data are included here to keep the charged-mode bookkeeping consistent with the row-level APS constraint $W_{\text{vac},\text{APS}}^0 = \frac{3}{2}N_a$ and to avoid double counting the identity representation.

There is one more row-level constraint on these Ramond weights that comes from the standard $\mathcal{N} = 2$ spectral-flow structure of the internal $c = 9$ SCFT. The following formula is the usual $\mathcal{N} = 2$ spectral-flow relation, imported here as part of the continuum lightcone CFT data:

$$h_\eta = h + \eta Q + \frac{c}{6}\eta^2, \quad Q_\eta = Q + \frac{c}{3}\eta, \quad c = 9, \quad (125)$$

where $\eta = \pm \frac{1}{2}$ maps NS states to R states, h is the internal conformal weight, and Q is the internal $U(1)$ charge before spectral flow. In the $\rho = 0$ spin-lift convention, the massless rows are assigned as

NS row	$\eta = +\frac{1}{2}$	$\eta = -\frac{1}{2}$
vac : $(h, Q, r) = (0, 0, 0)$	$S_+ : (h, Q, r) = (\frac{3}{8}, \frac{3}{2}, 0)$	$S_- : (h, Q, r) = (\frac{3}{8}, -\frac{3}{2}, 0)$
- : $(h, Q, r) = (\frac{1}{2}, -1, 2)$	$S_i^{(1)} : (h, Q, r) = (\frac{3}{8}, \frac{1}{2}, 2)$	not a ground-state row
+ : $(h, Q, r) = (\frac{1}{2}, 1, 1)$	not a ground-state row	$S_i^{(2)} : (h, Q, r) = (\frac{3}{8}, -\frac{1}{2}, 1)$

(126)

Here r is the $\mathbb{Z}_3$ charge, and the symbols $+$ and $-$ in the first column are the two adjacent massless NS rows, not signs of an orientifold eigenvalue. The entries marked “not a ground-state row” have $h_\eta > 3/8$ and therefore are not Ramond ground states.

Equation (126) is a bookkeeping constraint, not an orientifold projection. It says that the singlet Ramond weights w_+ and w_- belong to the same massless vac row as the NS identity operator. Therefore they must be reconciled with the APS identity-row constant $W_{\text{vac,APS}}^0 = \frac{3}{2}N_a$, rather than added as an independent extra identity-row subtraction. The triplet weights $w_i^{(1)}$ and $w_i^{(2)}$ belong to the two adjacent massless rows. In the standard four-DN vertex construction the adjacent NS representatives fail the $L_0 = 0$ test, but their Ramond spectral-flow partners can still enter the finite R zero-mode data. This is precisely why the Ramond row split in (99) has to be kept separate from the NS dimension count.

8.10 Ramond Weight Matching from APS Row Constants

The massless DN annulus blocks below give exact row-level constraints on the finite zero-mode weights. This subsection records only the constraints that follow from those row constants. It does not split the final $\vartheta_{10}(2\tau)q^{-1/4}$ and $\vartheta_{00}(2\tau)$ terms of (163) into NS and R pieces. In APS notation the labels (00), (01), (10), (11) refer to the four spin-structure traces before the spin-structure sum; the theta functions that remain after the sum are already recombined character expressions.

Let

$$W_{\text{vac,NS}}^0(a) = \omega_{\text{vac,NS}} N_a, \quad W_{\text{vac,R}}^0(a) = \omega_{\text{vac,R}} N_a \quad (127)$$

define the gauge-fixed NS and R identity-row constants per D7 Chan-Paton label. The row block (163) fixes only their sum:

$$\omega_{\text{vac,NS}} + \omega_{\text{vac,R}} = \frac{3}{2}. \quad (128)$$

The ordinary NS vertex-space dimension $\dim \mathcal{Z}_{\text{NS}}^{\text{proj}} = 2N_a$ does not by itself fix $\omega_{\text{vac,NS}}$, because W^0 is the $b_0 c_0$ gauge-fixed supertrace weight, not an ordinary dimension. A separate ghost-picture trace of the explicit identity-row representatives is needed to determine $\omega_{\text{vac,NS}}$ and $\omega_{\text{vac,R}}$ individually.

The adjacent rows are sharper. The standard four-DN NS vertex has no adjacent $L_0 = 0$ representatives, so the adjacent row constants in (173) must be reproduced by their Ramond spectral-flow partners. With the $\rho = 0$ row assignment (126), the APS row labels used in (166) impose

$$\sum_{i=1}^3 w_i^{(2)} = -\frac{3}{2}, \quad \sum_{i=1}^3 w_i^{(1)} = 0. \quad (129)$$

The first constraint is the $+$ row constant $-\frac{1}{2}M_+^{(a)} = -\frac{3}{2}N_{a-1}$ divided by N_{a-1} . The second constraint is the $-$ row constant 0 divided by N_{a+1} . If the bootstrap-fixed orientifold action and cocycles preserve the permutation symmetry of the three internal complex fermions, then

$$w_i^{(2)} = -\frac{1}{2}, \quad w_i^{(1)} = 0, \quad i = 1, 2, 3. \quad (130)$$

For an ordered $I \rightarrow S$ branch whose Ramond source states are the even branch $\chi_R^{IS} = 0$ in (107), the row-level Ramond weight has the form

$$W_R^0(a; 0, 0) = w_+ N_a - \frac{3}{2} N_{a-1}, \quad w_+ = \omega_{\text{vac}, R}. \quad (131)$$

The singlet parameter w_+ is fixed only after the identity-row split in (127) is fixed. The reverse ordered branch is represented in the ordinary charge-conjugating convention by $\chi_R^{SI} = 1$ and the target spaces in (109). If instead one declares the odd branch to be the ordered $I \rightarrow S$ branch, then the $+$ and $-$ row labels in the massless character formulas must be translated before using (129).

Combining (128) with (129), without choosing the individual identity-row split, gives the total massless zero-mode row constant

$$W_{\text{massless}}^{0, \text{row}}(a) = \frac{3}{2} N_a - \frac{3}{2} N_{a-1}. \quad (132)$$

This agrees with the large- t limit of the closed-form massless annulus (169). In the symmetric tadpole solution the signed constant (132) vanishes, but the ordinary vertex-space dimensions in (89) do not vanish. This is the precise sense in which the APS adjacent row trace is fixed while the identity-row NS/R split and the standalone K_0 question remain separate questions.

8.11 Low-lying Fractional Spectrum for $\mathbb{C}^3/\mathbb{Z}_3$

The projection formula immediately gives the low-lying fractional spectrum. This paragraph uses only the parent free-field action

$$g : \psi^i \mapsto \omega \psi^i, \quad g : \bar{\psi}^{\bar{i}} \mapsto \omega^{-1} \bar{\psi}^{\bar{i}}, \quad i = 1, 2, 3, \quad (133)$$

on the three complex internal fermions. The operator ψ^i has internal $U(1)$ charge $+1$, while $\bar{\psi}^{\bar{i}}$ has internal $U(1)$ charge -1 . Since $\mathcal{H}_{am}^{IS}$ is the ω^{a-m} eigenspace, the first ordered $I \rightarrow S$ open-string states are selected as follows, with all labels understood modulo 3:

parent internal operator	g -eigenvalue	fractional labels	comment
1	1	$m = a$	vacuum representation
$\psi^i, i = 1, 2, 3$	ω	$m = a - 1$	three chiral internal states
$\bar{\psi}^{\bar{i}}, \bar{i} = 1, 2, 3$	ω^{-1}	$m = a + 1$	three anti-chiral internal states

The first row is the vacuum representation contribution used in (157). The second and third rows are not ordinary massive (h, Q) representations. They are the local massless $(+)$ and $(-)$ representations of the extended $\mathcal{N} = 2$ algebra. Their presence means that the ED3-D7 sector contains charged zero modes or their superpartners. Therefore, for the standalone APS K_0 integral to be the correct object, the orientifold and Chan-Paton data must remove the corresponding $L_0 = 0$ vertex spaces. If they survive, the standalone K_0 formula is not applicable.

The reverse ordered sector $S \rightarrow I$ conjugates the g -eigenvalue and exchanges the chiral and anti-chiral rows. Thus a D7 stack with label $m = a - 1$ gives a chiral row in the $I \rightarrow S$ orientation and the conjugate anti-chiral row in the reverse orientation; similarly $m = a + 1$ gives the conjugate pair with the two orientations interchanged. The label $m = a$ gives the identity row in both orientations. This is the local BCFT zero-mode test.

For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$, every instanton label a couples before the orientifold projection to all three rows in the table: label a D7-branes give the vacuum row, label $a - 1$ D7-branes give the three chiral rows, and label $a + 1$ D7-branes give the three anti-chiral rows. Thus the symmetric local tadpole solution fails the standalone K_0 zero-mode test in the ordinary one-node projection.

8.12 Local Zero-mode Vector

This subsection assumes the involution-I geometry in which the space-filling D7-branes wrap the same internal exceptional divisor as the ED3, so that the parent internal ED3-D7 boundary condition contains the local identity row and its first fermionic descendants. For involution II, or for a D7 stack with a different internal boundary condition, one must replace (134) by the actual ED3-D7 spectrum.

It is useful to rewrite the preceding table as a vector. For a fixed instanton label a , define

$$\mu_{\text{vac}}^{\text{loc}}(a) = N_a, \quad \mu_+^{\text{loc}}(a) = 3N_{a-1}, \quad \mu_-^{\text{loc}}(a) = 3N_{a+1}, \quad (134)$$

where all labels are reduced modulo 3. The superscript “loc” means before applying the orientifold Chan-Paton projection. The number $\mu_{\text{vac}}^{\text{loc}}$ counts identity-row vacuum representations in the ordered $I \rightarrow S$ sector, while μ_+^{loc} and μ_-^{loc} count the chiral and anti-chiral massless rows. The factor 3 is the number of complex internal fermions ψ^i , $i = 1, 2, 3$, in (133).

Equivalently, the local adjacency matrices are

$$A_{am}^{\text{vac}} = \delta_{m,a}, \quad A_{am}^+ = 3\delta_{m,a-1}, \quad A_{am}^- = 3\delta_{m,a+1}, \quad (135)$$

and $\mu_\rho^{\text{loc}}(a) = \sum_m A_{am}^\rho N_m$ for $\rho \in \{\text{vac}, +, -\}$. Here “McKay-quiver bookkeeping” means the directed graph whose nodes are the three irreducible $\mathbb{Z}_3$ representations and whose arrows are generated by tensoring with the coordinate representation carried by the three complex internal fermions. Equations (134) and (135) are this bookkeeping specialized to the ED3-D7 open-string sector and to the orientation convention used in (13).

For a character lift with invariant instanton label $2a = s$, the orientifold maps the D7 label m to $2a - m$. Hence the fixed label $m = a$ maps to itself, while the adjacent labels $a - 1$ and $a + 1$ are exchanged.

We now spell out what the orientifold projection can do to these local charged rows. Let V_m be the Chan-Paton vector space of the label- m D7 stack, so that $\dim V_m = N_m$. In the ordered $I \rightarrow S$ orientation the local spaces counted by (134) are

$$W_{\text{vac}}^{IS}(a) = V_a, \quad W_+^{IS}(a) = \mathbb{C}^3 \otimes V_{a-1}, \quad W_-^{IS}(a) = (\mathbb{C}^3)^\vee \otimes V_{a+1}. \quad (136)$$

The factor $\mathbb{C}^3$ is spanned by the three operators ψ^i , and $(\mathbb{C}^3)^\vee$ is spanned by the conjugate operators $\bar{\psi}^i$. The reverse orientation has the dual Chan-Paton spaces and is included in the trace convention of (148).

The orientifold maps an ordered $I \rightarrow S$ state to an ordered $S \rightarrow I$ state. Abstractly, on a pair of spaces W and W' of equal dimension it acts by an off-diagonal involution

$$\Theta(w, w') = (\epsilon T^{-1} w', \epsilon T w), \quad T : W \longrightarrow W', \quad \epsilon = \pm 1, \quad (137)$$

where T contains the local CFT action and the Chan-Paton matrices. The $+1$ eigenspace has dimension $\dim W$, because it is parametrized by $w \in W$ with $w' = \epsilon T w$. Thus the orientation-pair projection identifies the two orientations; it does not by itself remove a nonzero charged row. It can change the reality condition and gauge representation of the surviving variables, but not their complex character count.

Consequently, for a one-node invariant instanton and an ordinary orientation-pair projection, the local projected counts are

$$\mu_{\text{vac}}^{\text{proj}}(a) = N_a, \quad \mu_{\text{adj}}^{\text{proj}}(a) = 3N_{a-1} = 3N_{a+1}, \quad (138)$$

where the equality $N_{a-1} = N_{a+1}$ is the D7 image constraint for the lift $s = 2a$. The symbol $\mu_{\text{adj}}^{\text{proj}}$ counts the single orientifold-invariant adjacent pair descending from the local $(+)$ and $(-)$ rows. If a more exotic Chan-Paton implementation is proposed, it must be given explicitly as a replacement for (137).

Proposition 2 (Local obstruction to a standalone K_0 number). *Assume the local ED3-D7 spectrum (136), a one-node invariant instanton label a with character lift $s = 2a$, nonnegative D7 multiplicities N_m , and the ordinary orientation-pair projection (137). Then the local projected ED3-D7 zero-mode test is satisfied if and only if*

$$N_0 = N_1 = N_2 = 0. \quad (139)$$

Proof. The ordinary projected counts are (138). The condition $\mu_{\text{vac}}^{\text{proj}}(a) = 0$ gives $N_a = 0$, and the adjacent-pair condition $\mu_{\text{adj}}^{\text{proj}}(a) = 0$ gives $N_{a-1} = 0$. The D7 image constraint for the character lift $s = 2a$ is $N_{a-1} = N_{a+1}$, so $N_{a+1} = 0$ as well. These three labels exhaust the fractional D7 nodes. The converse is immediate. $\square$

Thus, in the local involution-I sector, any nonzero tadpole-cancelling D7 stack produces projected ED3-D7 charged modes under the ordinary orientifold projection. In that case the standalone K_0 integral (183) is not the applicable object.

Corollary 1 (Untwisted O7 charge obstructs the standalone K_0 formula). *Assume the hypotheses of Proposition 2: the local ED3-D7 spectrum (136), a one-node invariant instanton, nonnegative D7 multiplicities, and the ordinary orientation-pair projection. If the tadpole equations are solved with an untwisted crosscap coefficient $C_0 < 0$, then the standalone K_0 zero-mode test fails. In particular, for the involution-I O7 $^-$ normalization $C_0 = -8\sqrt{3}$, no solution of the local tadpole equations can give a standalone superpotential prefactor K_0 under these assumptions.*

Proof. If the standalone- K_0 zero-mode test passed, Proposition 2 would give $N_0 = N_1 = N_2 = 0$. But any nonnegative tadpole solution satisfies (35). If $C_0 < 0$, that equation gives $\sum_m N_m = -\sqrt{3}C_0 > 0$, contradiction. The O7 $^-$ value $C_0 = -8\sqrt{3}$ gives (36), hence the claimed special case. $\square$

The necessary local test for a standalone O(1) superpotential prefactor is

$$\mu_{\text{vac}}^{\text{proj}}(a) = \mu_+^{\text{proj}}(a) = \mu_-^{\text{proj}}(a) = 0, \quad (140)$$

where “proj” denotes the spectrum after the orientifold and Chan-Paton projection. If the orientifold exchanges the $(+)$ and $(-)$ rows, the last two equalities mean the single condition $\mu_{\text{adj}}^{\text{proj}}(a) = 0$. For the ordinary orientation-pair projection (138), this condition becomes

$$N_a = N_{a-1} = N_{a+1} = 0. \quad (141)$$

Since the three labels exhaust the local $\mathbb{Z}_3$ fractional branes, (141) means that no local D7 stack intersects the instanton in this BCFT sector. Thus a local tadpole solution with any D7 stack present fails the standalone K_0 zero-mode test under the ordinary orientation-pair projection. A different conclusion would require an explicitly specified orientifold and Chan-Paton action replacing (137).

9 Mobius Strip Contribution

The Mobius strip has one boundary on the instanton and one crosscap. It is the unoriented one-boundary trace with an orientifold insertion; it is not a second annulus projection. Define

$$\hat{\tau} = \tau + \frac{1}{2}, \quad \hat{q} = e^{2\pi i \hat{\tau}} = -e^{-2\pi t}, \quad \tau = it. \quad (142)$$

For a representation (h, Q) , the orientifold action contributes a phase

$$\alpha_{h,Q,\sigma} = \sigma e^{-i\pi(h-Q/2)}, \quad \sigma = \pm 1. \quad (143)$$

The Mobius partition function is

$$Z_M(t) = \sum_{Q=0}^1 (-1)^{1-Q} \vartheta_{1-Q,0}(2\hat{\tau}) \sum_{h \geq |Q|/2} \sum_{\sigma=\pm 1} \sigma \tilde{n}_{h,Q,\sigma} e^{-i\pi(h-Q/2)} \hat{q}^{h-(1+Q)/4}. \quad (144)$$

The integer $\tilde{n}_{h,Q,\sigma}$ is the signed character coefficient of representations of type (h, Q) in the instanton-instanton open-string spectrum with orientifold sign σ . It is an ordinary nonnegative multiplicity only for genuine massive representations.

For the $O(1)$ instanton convention, the vacuum representation has

$$\sigma_{\text{vac}} = -1. \quad (145)$$

This sign is fixed by requiring the four translation zero modes to be orientifold even while the ghost zero modes are projected out. Non-vacuum signs are local CFT data and must be computed from the action of Ω_σ on the corresponding fractional open-string states.

In the same massive-character basis used above, the vacuum representation gives

$$\tilde{n}_{0,0,-1} = 1, \quad \tilde{n}_{1/2,1,-1} = -1, \quad \tilde{n}_{0,0,+1} = \tilde{n}_{1/2,1,+1} = 0. \quad (146)$$

Substitution into (144) gives the universal vacuum contribution

$$Z_M^{\text{vac}}(t) = \vartheta_{10}(2\hat{\tau}) \hat{q}^{-1/4} + \vartheta_{00}(2\hat{\tau}). \quad (147)$$

As $t \rightarrow \infty$, this tends to 3, the net universal zero-mode weight that is removed by the last term in (183).

10 Meaning of Z_A and Z_M

The notation Z_A and Z_M in the D-instanton reference is easy to misread. In this note we use the following convention.

Definition 2 (Annulus and Mobius integrands). *The symbols $Z_A(t)$ and $Z_M(t)$ denote the open-channel integrands that appear in the APS one-loop exponent with measure $dt/(2t)$. They are not the already integrated annulus and Mobius amplitudes.*

One precise trace-level version of this convention is the following. Let $P_{\text{GSO}} = (1 + (-1)^f)/2$, where $(-1)^f$ is the worldsheet GSO parity. Let STr^{gf} denote the supertrace over the gauge-fixed open-string Fock space; equivalently it includes the spacetime-statistics sign and the signs from the b, c, β, γ ghost oscillators. If one traces over the unreduced ghost Fock space, these ghost signs may be written explicitly as a factor such as $(-1)^{N_{bc} + N_{\beta\gamma}}$. In the character formulas above they have already been included in the free-field blocks $f_{\alpha\beta}$.

For a single invariant instanton boundary condition I and the total space-filling boundary condition $S = \oplus_m N_m \tilde{S}_m$, the open-channel traces may be written as

$$Z_A(t) = \frac{1}{2} \text{STr}_{\mathcal{H}_{II} \oplus \mathcal{H}_{IS} \oplus \mathcal{H}_{SI}}^{\text{gf}} \left(P_{\text{GSO}} b_0 c_0 e^{-2\pi t L_0} \right), \quad (148)$$

$$Z_M(t) = \frac{1}{2} \text{STr}_{\mathcal{H}_{II}}^{\text{gf}} \left(\Omega_\sigma P_{\text{GSO}} b_0 c_0 e^{-2\pi t L_0} \right). \quad (149)$$

Here $\mathcal{H}_{IS} = \oplus_m N_m \mathcal{H}_{am}^{IS}$ and $\mathcal{H}_{SI} = \oplus_m N_m \mathcal{H}_{ma}^{SI}$ include both ordered ED3-D7 orientations. The leading factor $1/2$ is the orientifold group average; the Ω_σ part of the average is the Mobius trace, while the identity part is the annulus trace. If instead one defines the Hilbert spaces to be already orientifold-projected, this factor is absorbed into that definition. One must not use both conventions at once. The operator $b_0 c_0$ is a projector onto the Siegel-gauge representatives because $\{b_0, c_0\} = 1$ and b_0 annihilates those representatives.

The annulus integrand $Z_A(t)$ is a sum of two types of annuli:

$$Z_A(t) = Z_{A,\text{DD}}(t) + Z_{A,\text{DN}}(t). \quad (150)$$

Here $Z_{A,\text{DD}}$ has both boundaries on the instanton, while $Z_{A,\text{DN}}$ has one boundary on the instanton and one boundary on the tadpole-cancelling space-filling branes. The symbol “DN” in $Z_{A,\text{DN}}$ refers to the four non-compact directions, as explained before (41); the internal fractional data are contained in the signed coefficients $n_{h,Q}$.

The Mobius integrand $Z_M(t)$ has one boundary on the instanton and one crosscap. It is not a separate projection of $Z_A(t)$. It is the contribution of the orientifold insertion in the one-boundary open-string trace, equivalently the instanton-crosscap overlap in the closed channel.

The closed-channel cutoffs are not identical for annulus and Mobius. If ℓ is the closed-channel Schwinger parameter, then

$$\ell = \frac{1}{t} \quad \text{for annuli}, \quad \ell = \frac{1}{4t} \quad \text{for the Mobius strip}. \quad (151)$$

This is why the reference formula uses a lower cutoff ε' for the annulus integral but $\varepsilon'/4$ for the Mobius integral. Dropping this distinction changes the UV subtraction in the open channel.

Finally, the calligraphic closed-channel quantities, often written as $\mathcal{Z}_A$ and $\mathcal{Z}_M$ in derivations, are the same physical contributions expressed after modular transformation. They should not be added to the open-channel Z_A and Z_M as extra terms.

11 Symmetric Tadpole, DN Annulus, and Zero-mode Test

11.1 Tadpole Solution Under the Symmetric Crosscap Assumption

Assume involution I with the APS O7⁻ orientifold (14). In the Ishibashi normalization of (24), assume

$$C_0 = -8\sqrt{3}, \quad C_1 = C_2 = 0. \quad (152)$$

This is the precise version of the statement that the crosscap has untwisted D7 charge -8 in bulk D7 units and no twisted RR charge in this local basis. Then (24) is solved by

$$N_0 = N_1 = N_2 = 8. \quad (153)$$

Indeed, for $\ell = 0$ the D7 coefficient is

$$\frac{1}{\sqrt{3}} \sum_{m=0}^2 N_m = \frac{24}{\sqrt{3}} = 8\sqrt{3}, \quad (154)$$

which cancels $C_0 = -8\sqrt{3}$. For $\ell = 1, 2$ the D7 coefficient is

$$\frac{8}{\sqrt{3}} (1 + \omega^\ell + \omega^{2\ell}) = 0, \quad (155)$$

which cancels the assumed $C_1 = C_2 = 0$.

This result depends on the stated crosscap assumption. If the direct crosscap computation gives nonzero C_1 or C_2 , then (153) is not the tadpole solution.

11.2 Exact Annulus Coefficients for a Fractional Instanton

Fix the fractional instanton label a . If the orientifold and Chan-Paton projection remove the local (+) and (−) massless rows in the table above, the DN annulus integrand in the open channel is

$$Z_{A,\text{DN}}^{(a)}(\tau) = \sum_{Q=0}^1 \frac{(-1)^Q}{2} \vartheta_{1-Q,0}(2\tau) \sum_{h \geq |Q|/2} \left(\sum_{m=0}^2 N_m n_{h,Q}^{(a,m)} \right) q^{h-(1+Q)/4}, \quad (156)$$

with $n_{h,Q}^{(a,m)}$ given by the projection formula (44). This is the corrected replacement for the earlier expression with a double sum over two D7 labels. There is only one D7 boundary on this annulus, so the dependence on D7 multiplicities is linear.

If the (+) or (−) rows survive, (156) is only the part captured by the massive-character coefficients. The surviving massless rows must be kept separately as charged zero-mode data.

Suppose, as an additional assumption, that the internal ED3-D7 boundary conditions coincide and therefore the parent internal spectrum contains the identity boundary operator. The identity has g -eigenvalue 1, so it contributes only when the D7 label equals the instanton label:

$$n_{0,0}^{(a,m),\text{id}} = \delta_{am}, \quad n_{1/2,1}^{(a,m),\text{id}} = -\delta_{am}. \quad (157)$$

Substituting (157) into (156) gives the identity contribution

$$Z_{A,\text{DN}}^{(a),\text{id}}(\tau) = \frac{N_a}{2} \left[\vartheta_{10}(2\tau) q^{-1/4} + \vartheta_{00}(2\tau) \right]. \quad (158)$$

For the symmetric tadpole solution (153), this is

$$Z_{A,\text{DN}}^{(a),\text{id}}(\tau) = 4 \left[\vartheta_{10}(2\tau) q^{-1/4} + \vartheta_{00}(2\tau) \right]. \quad (159)$$

This formula is often the tempting “explicit annulus” written in rough notes. Its correct interpretation is narrower: it is only the identity part of the internal ED3-D7 spectrum, not the full annulus determinant.

11.3 Massless DN Rows

The corrected massive-character formula (156) is not, by itself, the complete local DN annulus when the massless rows survive. In the parent-boundary convention of Assumption 4, the low-lying charge-sector table (52) gives the following exact row multiplicities for a fractional instanton of label a :

$$M_{\text{vac}}^{(a)} = N_a, \quad M_+^{(a)} = 3N_{a-1}, \quad M_-^{(a)} = 3N_{a+1}. \quad (160)$$

Here $M_\lambda^{(a)}$ is an annulus row multiplicity before deleting charged zero-mode representatives, and the labels $a \pm 1$ are read in $\mathbb{Z}/3\mathbb{Z}$. For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$,

$$M_{\text{vac}}^{(a)} = 8, \quad M_+^{(a)} = 24, \quad M_-^{(a)} = 24. \quad (161)$$

Let $\mathfrak{Z}_{\lambda, \text{DN}}(t)$, with $\lambda \in \{\text{vac}, +, -\}$, denote the DN annulus block obtained by inserting the internal massless character λ into the APS spin-structure sum with the four non-compact DN free-field factor. This definition fixes a function of t ; it is not a dimension count. In this notation the massless-row part of the DN annulus is

$$Z_{A, \text{DN}}^{(a), \text{massless}}(t) = M_{\text{vac}}^{(a)} \mathfrak{Z}_{\text{vac}, \text{DN}}(t) + M_+^{(a)} \mathfrak{Z}_{+, \text{DN}}(t) + M_-^{(a)} \mathfrak{Z}_{-, \text{DN}}(t). \quad (162)$$

The identity-row block is fixed explicitly by (158):

$$\mathfrak{Z}_{\text{vac}, \text{DN}}(t) = \frac{1}{2} \left[\vartheta_{10}(2\tau) q^{-1/4} + \vartheta_{00}(2\tau) \right], \quad \tau = it, \quad q = e^{-2\pi t}. \quad (163)$$

The adjacent-row blocks can also be evaluated from the APS massless character formula. This is a continuum lightcone-string character input, not a derivation from the finite orbifold projection. In APS notation,

$$g^{(\pm)}(z, \tau) = \pm \frac{1}{2} [f_{3,1}(z, \tau) - f_{3,-1}(z, \tau)] + \frac{1}{2} g\left(z, \tau; \frac{1}{2}, 1\right), \quad (164)$$

where $g(z, \tau; h, Q)$ is the massive character seed used in (41). The difference $f_{3,1} - f_{3,-1}$ vanishes at $z = 0, \frac{1}{2}, \frac{\tau}{2}$, but at $z = \frac{1+\tau}{2}$ it gives

$$g_{11}^{(+)} = +1, \quad g_{11}^{(-)} = -1. \quad (165)$$

The subscript 11 denotes the R sector with $(-1)^f$ insertion in the APS spin-structure sum. For the four-DN free-field factor, the same APS convention has $2f_{11}^{\text{DN}} = -1$, hence $f_{11}^{\text{DN}} = -\frac{1}{2}$ in the sum $\frac{1}{2} \sum_{\alpha, \beta} f_{\alpha\beta}^{\text{DN}} g_{\alpha\beta}$. Combining this odd spin-structure term with the massive seed contribution

$$\mathfrak{Z}_{h=1/2, Q=1; \text{DN}}(t) = -\frac{1}{2} \vartheta_{00}(2\tau),$$

one obtains the exact adjacent massless-row DN blocks

$$\mathfrak{Z}_{+, \text{DN}}(t) = -\frac{1}{4} \vartheta_{00}(2\tau) - \frac{1}{4}, \quad \mathfrak{Z}_{-, \text{DN}}(t) = -\frac{1}{4} \vartheta_{00}(2\tau) + \frac{1}{4}. \quad (166)$$

Thus their large- t constants are

$$\lim_{t \rightarrow \infty} \mathfrak{Z}_{\text{vac}, \text{DN}}(t) = \frac{3}{2}, \quad \lim_{t \rightarrow \infty} \mathfrak{Z}_{+, \text{DN}}(t) = -\frac{1}{2}, \quad \lim_{t \rightarrow \infty} \mathfrak{Z}_{-, \text{DN}}(t) = 0. \quad (167)$$

For the symmetric tadpole solution, the signed large- t constant of the full massless DN annulus is therefore

$$8 \cdot \frac{3}{2} + 24 \cdot \left(-\frac{1}{2}\right) + 24 \cdot 0 = 0. \quad (168)$$

More generally, substituting (163) and (166) into (162) gives

$$Z_{A,\text{DN}}^{(a),\text{massless}}(t) = \frac{N_a}{2} \vartheta_{10}(2\tau) q^{-1/4} + \left[\frac{N_a}{2} - \frac{3}{4} (N_{a-1} + N_{a+1}) \right] \vartheta_{00}(2\tau) + \frac{3}{4} (N_{a+1} - N_{a-1}). \quad (169)$$

If the D7 image constraint for a one-node invariant instanton gives $N_{a-1} = N_{a+1}$, the last constant term in (169) vanishes. For the symmetric solution this reduces to

$$Z_{A,\text{DN}}^{(a),\text{massless}}(t) = 4 \vartheta_{10}(2\tau) q^{-1/4} - 8 \vartheta_{00}(2\tau), \quad (170)$$

whose large- t limit is zero because $\vartheta_{10}(2it) q^{-1/4} \rightarrow 2$ and $\vartheta_{00}(2it) \rightarrow 1$. This cancellation is a supertrace statement. It does not mean that the charged $L_0 = 0$ states vanish. The ordinary vertex-space dimensions in (89) still show nonzero charged bosonic and fermionic zero-mode spaces. Equation (168) only says that their gauge-fixed signed weights can cancel in the annulus integrand.

The determinant keeps the nonzero descendants of a massless row but removes the actual $L_0 = 0$ zero-mode representatives. Therefore the massless contribution to the determinant has the form

$$Z_{A,\text{DN}}^{(a),\text{massless},\text{det}}(t) = \sum_{\lambda \in \{\text{vac}, +, -\}} M_{\lambda}^{(a)} \mathfrak{Z}_{\lambda,\text{DN}}(t) - \sum_{\lambda \in \{\text{vac}, +, -\}} W_{\lambda}^0(a). \quad (171)$$

The second term in (171) is a constant in t because it is the contribution of $L_0 = 0$ states to the trace. It removes zero-mode representatives, not entire massless characters. The identity-row subtraction is fixed:

$$W_{\text{vac}}^0(a) = W_{\text{vac},\text{APS}}^0(a) = \frac{3}{2} N_a. \quad (172)$$

The adjacent-row constants in the same APS row-level trace convention are constrained by (167):

$$W_{+,\text{row}}^0(a) = -\frac{1}{2} M_+^{(a)} = -\frac{3}{2} N_{a-1}, \quad W_{-,\text{row}}^0(a) = 0. \quad (173)$$

Here the subscript “row” reminds the reader that this is the signed gauge-fixed row constant seen by the annulus character, not an ordinary dimension. The standard four-DN NS representatives in the adjacent rows do not contribute $L_0 = 0$ zero modes, but the Ramond spectral-flow partners in (126) can contribute. Therefore the finite Ramond weights in (122)–(123), with the fixed R GSO branch and orientifold pairing, must reproduce the row constants in (173). This is a nontrivial consistency check on the finite zero-mode BCFT.

Combining the massive and massless pieces, the DN part of the determinant is

$$Z_{A,\text{DN}}^{(a),\text{det}}(t) = Z_{A,\text{DN}}^{(a),\text{massive}}(t) + Z_{A,\text{DN}}^{(a),\text{massless},\text{det}}(t), \quad (174)$$

where $Z_{A,\text{DN}}^{(a),\text{massive}}$ is computed from the massive coefficients $d_{h,Q;r}^{IS}$ by (49) and (156), excluding the massless rows $\text{vac}, +, -$. A necessary bookkeeping check is

$$\lim_{t \rightarrow \infty} Z_{A,\text{DN}}^{(a),\text{det}}(t) = 0. \quad (175)$$

If (175) fails, then either a charged zero mode has not been removed, or a nonzero-mode descendant has been incorrectly deleted with an entire massless row.

11.4 Charged Zero-mode Test

The identity row in (134) already gives a sharp obstruction. The large- t limit of (158) is

$$Z_{A,\text{DN}}^{(a),\text{id}}(t) \longrightarrow \frac{3}{2}N_a, \quad t \rightarrow \infty. \quad (176)$$

Here $\vartheta_{10}(2it)q^{-1/4} \rightarrow 2$ and $\vartheta_{00}(2it) \rightarrow 1$. Therefore the identity contribution represents additional ED3-D7 zero modes. Under the assumptions of the APS one-loop normalization formula for a standalone superpotential term, the only zero modes are the four translations and two Goldstinos. Thus a standalone K_0 computation requires

$$\mu_{\text{vac}}^{\text{proj}}(a) = \mu_+^{\text{proj}}(a) = \mu_-^{\text{proj}}(a) = 0. \quad (177)$$

This is the local representation-theoretic statement; if the orientifold exchanges the (+) and (−) rows, it is read as the vanishing of the adjacent-pair invariant subspace. At the trace level, vanishing of the signed large- t constant is only a necessary determinant check:

$$\mathfrak{M}_{\text{ch}} = 0 \implies \lim_{t \rightarrow \infty} Z_{A,\text{DN}}(t) = 0, \quad (178)$$

but the converse is false. The symmetric cancellation (168) is the local example: the annulus supertrace constant cancels even though the charged vertex spaces are nonzero. If (176) is nonzero, the standalone K_0 formula (183) does not apply.

For the symmetric solution (153), $N_a = 8$ for every fractional label a . Hence the coincident-internal-boundary assumption leading to (159) is incompatible with the standalone- K_0 zero-mode assumption. Moreover

$$(\mu_{\text{vac}}^{\text{loc}}, \mu_+^{\text{loc}}, \mu_-^{\text{loc}}) = (8, 24, 24) \quad (179)$$

for every instanton label before orientifold projection. This row-level representation vector is defined before the $L_0 = 0$ vertex representatives are selected. Under the ordinary orientation-pair projection (138), this descends to

$$(\mu_{\text{vac}}^{\text{proj}}, \mu_{\text{adj}}^{\text{proj}}) = (8, 24). \quad (180)$$

Thus the adjacent-label rows are not removed; their two orientations are identified. This is the main local conclusion for the ordinary orientation-pair projection. After the external DN weight test, the actual standard NS charged zero-mode count is instead (89): the identity row gives 16 projected bosonic charged zero modes for $N_a = 8$, while the adjacent rows are not NS $L_0 = 0$ representatives unless additional boundary data changes the vertex construction.

11.5 Mobius Vacuum and the Incorrect Crosscap Multiplier

The universal Mobius vacuum contribution is (147), not an O7 charge multiplier times the DN annulus block. Its large- t limit is

$$Z_M^{\text{vac}}(t) \longrightarrow 3. \quad (181)$$

The O7 charge enters the closed-channel tadpole coefficients C_ℓ and the non-vacuum orientifold data $\tilde{n}_{h,Q,\sigma}$; it is not a scalar factor $M = -8$ multiplying the open-channel vacuum character. With no extra DN zero modes, the combination

$$Z_M^{\text{vac}}(t) + 3 \left(e^{-2\pi h_{\text{reg}} t} - 1 \right) \longrightarrow 0 \quad (t \rightarrow \infty) \quad (182)$$

is exactly the APS zero-mode cancellation.

12 The One-loop Normalization Formula

Let $Z_A(t)$ and $Z_M(t)$ be the open-channel integrands defined in (150) and (144). The contribution $Z_{A,DD}$ vanishes representation by representation under the assumptions above, but its zero-mode regularization still contributes to the universal normalization.

The formula below is imported from Alexandrov–Pioline–Sen. Its input is the BCFT open-channel data $Z_A(t)$ and $Z_M(t)$; the task of this note is to compute these data for the local $\mathbb{C}^3/\mathbb{Z}_3$ orientifold and then check whether the zero-mode assumptions of the formula hold.

The one-loop normalization in the reference convention is

$$K_0 = h_{\text{reg}}^{3/2} \lim_{\varepsilon \rightarrow 0} \lim_{\varepsilon' \rightarrow 0} \exp \left[\int_{\varepsilon'}^{1/\varepsilon} \frac{dt}{2t} Z_A(t) + \int_{\varepsilon'/4}^{1/\varepsilon} \frac{dt}{2t} Z_M(t) + 3 \int_0^{1/\varepsilon} \frac{dt}{2t} \left(e^{-2\pi h_{\text{reg}} t} - 1 \right) \right]. \quad (183)$$

Here $h_{\text{reg}} > 0$ is the auxiliary conformal weight used to regulate the universal zero-mode contribution. The final result is independent of h_{reg} when the reference assumptions hold.

The coefficient 3 in (183) is not the number of bosonic zero modes. It is the net zero-mode regularization weight obtained from four bosonic translation modes and two fermionic Goldstino modes in the $O(1)$ instanton normalization. Within the APS normalization (183), replacing this coefficient by 4 is an error: it would misregularize the universal zero-mode contribution and spoil the cancellation with the prefactor $h_{\text{reg}}^{3/2}$.

13 Exact Open-channel Determinant Computation

The determinant integral is evaluated after the $L_0 = 0$ representatives have been removed. In the present parent-boundary convention the ED3-D7 annulus is computed directly from the $\mathbb{C}^3/\mathbb{Z}_3$ oscillator generating function: subtract the three massless $\mathcal{N} = 2$ rows, triangularly extract the massive extended-character multiplicities, and resum the result to the A_2 -theta expression. The Mobius strip is then computed from the instanton-instanton orientifold action. This section uses q -series as an exact coefficient language; the final annulus and Mobius finite parts are not approximations.

13.1 Positive-energy Coefficient Convention

Throughout this section “det” means that the zero-mode representatives have been removed from the trace. The coefficients c_Δ below are signed determinant coefficients at positive open-string energy $\Delta > 0$; they are not dimensions of vertex spaces.

For the positive-energy string-mode series, separate the exponentially decaying regulator term in (183) and define

$$\mathcal{I}_{\text{open}}^{\text{nz}}(t) = Z_A^{\text{det}}(t) + Z_M^{\text{det}}(t) - 3. \quad (184)$$

Here Z_A^{det} is the annulus integrand with its $L_0 = 0$ representatives removed; its DN part is organized as in (174). The Mobius determinant integrand is $Z_M^{\text{det}} = Z_M$ with the ordinary $O(1)$ neutral zero-mode convention of (147). The constant -3 removes the universal large- t neutral zero-mode contribution that is paired with $+3e^{-2\pi h_{\text{reg}} t}$ in (183). When additional neutral zero modes are present, $\mathcal{I}_{\text{open}}^{\text{nz}}$ is only the determinant supertrace; a standalone K_0 also requires the zero-mode spectrum to reduce to the universal $O(1)$ sector. For large t , expand

$$\mathcal{I}_{\text{open}}^{\text{nz}}(t) = \sum_{\Delta > 0} c_\Delta \exp(-2\pi t \Delta), \quad (185)$$

where Δ is the open-string L_0 eigenvalue in the chosen convention and c_Δ is the signed gauge-fixed degeneracy including annulus and Mobius contributions. The condition $\Delta > 0$ is part of the definition: the zero-mode constants have already been subtracted. The remaining $+3e^{-2\pi h_{\text{reg}} t}$ term in (183) is the APS regulator term, not a BCFT degeneracy coefficient. For any finite coefficient list, the large- t contribution from $t \geq t_0 > 0$ is

$$\int_{t_0}^{\infty} \frac{dt}{2t} \sum_{\Delta \leq \Delta_{\text{max}}} c_\Delta e^{-2\pi t \Delta} = \frac{1}{2} \sum_{\Delta \leq \Delta_{\text{max}}} c_\Delta E_1(2\pi \Delta t_0), \quad E_1(x) = \int_x^{\infty} \frac{e^{-u}}{u} du. \quad (186)$$

The small- t endpoint is handled below by exact modular transformations, first for the A_2 annulus theta function and then for the folded Mobius product.

13.2 DN Annulus Coefficients for the q -series

This subsection spells out the part of the coefficients c_Δ in (185) that comes from the ED3-D7 annulus. The Mobius coefficients $\tilde{n}_{h,Q,\sigma}$ are added in the Mobius subsections below.

First consider the massless rows. Define

$$A_a = \frac{1}{2}N_a, \quad B_a = \frac{1}{2}N_a - \frac{3}{4}(N_{a-1} + N_{a+1}), \quad W_a^{\text{row}} = \frac{3}{2}(N_a - N_{a-1}), \quad (187)$$

where all D7 labels are read in $\mathbb{Z}/3\mathbb{Z}$. The constant W_a^{row} is the total massless row subtraction in the APS row-level trace convention:

$$W_a^{\text{row}} = W_{\text{vac}}^0(a) + W_{+, \text{row}}^0(a) + W_{-, \text{row}}^0(a) = \frac{3}{2}N_a - \frac{3}{2}N_{a-1}. \quad (188)$$

Using

$$\vartheta_{10}(2it)q^{-1/4} = \sum_{n \in \mathbb{Z}} q^{n(n+1)}, \quad \vartheta_{00}(2it) = \sum_{n \in \mathbb{Z}} q^{n^2}, \quad q = e^{-2\pi t}, \quad (189)$$

the exact massless determinant contribution (171) becomes

$$Z_{A, \text{DN}}^{(a), \text{massless}, \text{det}}(t) = A_a \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0, -1}} q^{n(n+1)} + B_a \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} q^{n^2}. \quad (190)$$

The omitted terms are precisely the q^0 terms. Indeed, $2A_a + B_a - W_a^{\text{row}} + \frac{3}{4}(N_{a+1} - N_{a-1}) = 0$, which is the large- t zero-mode subtraction check in coefficient form. Thus (190) contains only positive open-string energies. For the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$,

$$A_a = 4, \quad B_a = -8, \quad W_a^{\text{row}} = 0, \quad (191)$$

so the massless descendant contribution is

$$Z_{A, \text{DN}}^{(a), \text{massless}, \text{det}}(t) = 4 \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0, -1}} q^{n(n+1)} - 8 \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} q^{n^2}. \quad (192)$$

This is an exact annulus statement in the APS convention. The negative coefficients are signed supertrace coefficients; they are not negative dimensions of Hilbert spaces. In the notation of (185), $c_\Delta^{A, \text{DN}, \text{massless}}(a)$ is the coefficient of q^Δ in (190). Equivalently, for $\Delta > 0$ define

$$\mathfrak{t}(\Delta) = \# \{n \in \mathbb{Z} \mid n \neq 0, -1, \quad \Delta = n(n+1)\}, \quad \mathfrak{s}(\Delta) = \# \{n \in \mathbb{Z} \mid n \neq 0, \quad \Delta = n^2\}. \quad (193)$$

Then

$$c_{\Delta}^{A,\text{DN},\text{massless}}(a) = A_a \mathfrak{t}(\Delta) + B_a \mathfrak{s}(\Delta). \quad (194)$$

This is an exact coefficient formula for the massless descendant part. It applies only to the annulus DN massless rows; massive representations and Mobius signs are added separately below.

For the massive rows, fix a truncation $\Delta_{\max} > 0$. Let $\mathcal{K}_{\Delta_{\max}}$ be the finite list of massive extended $\mathcal{N} = 2$ characters $\chi_{\kappa,\text{NS}}$, with $\kappa = (h, Q)$, whose DN block can contribute to (185) with $\Delta \leq \Delta_{\max}$. Order this list by increasing leading NS character power. Write

$$\chi_{\kappa,\text{NS}}(u; q) = b_{\kappa} q^{e_{\kappa}} u^{J_{\kappa}} + \chi_{\kappa,\text{NS}}^{>}(u; q), \quad b_{\kappa} \neq 0. \quad (195)$$

Here e_{κ} , J_{κ} , and b_{κ} are fixed by the character normalization. The remainder $\chi_{\kappa,\text{NS}}^{>}$ contains only terms with higher q -power, or terms at the same q -power that occur later in the triangular monomial order. If two characters begin at the same q -power, fix an ordering of their independent u -monomials that makes the finite coefficient matrix triangular. This is a basis choice in the finite character-extraction problem, not a physical assumption.

For each $\ell = 0, 1, 2$, set

$$R_0^{(\ell)}(u; q) = \mathcal{P}_{\text{NS}}^{(\epsilon_{\text{NS}})}(\omega^{\ell}, u; q) - \sum_{\lambda \in \{\text{vac}, +, -\}} A_{\lambda}^{IS}(\ell) \chi_{\lambda,\text{NS}}(u; q), \quad (196)$$

where the massless coefficients $A_{\lambda}^{IS}(\ell)$ are fixed by (51). Suppose $\mathcal{K}_{\Delta_{\max}} = \{\kappa_1, \dots, \kappa_L\}$ in the chosen triangular order. The massive twined traces are extracted recursively by

$$T_{\kappa_j}^{IS}(\ell) = \frac{[q^{e_{\kappa_j}} u^{J_{\kappa_j}}] R_{j-1}^{(\ell)}(u; q)}{b_{\kappa_j}}, \quad (197)$$

$$R_j^{(\ell)}(u; q) = R_{j-1}^{(\ell)}(u; q) - T_{\kappa_j}^{IS}(\ell) \chi_{\kappa_j,\text{NS}}(u; q), \quad j = 1, \dots, L. \quad (198)$$

All series in (196)–(198) are truncated only after all terms that can influence $\Delta \leq \Delta_{\max}$ have been kept. Therefore the extracted $T_{\kappa_j}^{IS}(\ell)$ are exact below the cutoff, while representations whose first possible contribution has $\Delta > \Delta_{\max}$ are part of the large- t open-series tail beyond the cutoff.

Fourier transforming the result gives

$$d_{\kappa;r}^{IS} = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-r\ell} T_{\kappa}^{IS}(\ell), \quad n_{\kappa}^{(a,m)} = d_{\kappa;a-m}^{IS}. \quad (199)$$

For the annulus between the label- a ED3 and the full D7 stack, define

$$N_{\kappa}^{(a)} = \sum_{m=0}^2 N_m d_{\kappa;a-m}^{IS}. \quad (200)$$

If $N_0 = N_1 = N_2 = 8$, then $N_{\kappa}^{(a)} = 8T_{\kappa}^{IS}(0)$, as in (50).

The massive DN annulus contribution to the open-channel coefficients is now explicit. For $\kappa = (h, Q)$, let

$$\delta_Q(n) = \begin{cases} \left(n + \frac{1}{2}\right)^2, & Q = 0, \\ n^2, & Q = 1, \end{cases} \quad n \in \mathbb{Z}. \quad (201)$$

Then the massive DN coefficient at open energy $\Delta > 0$ is

$$c_{\Delta}^{A,\text{DN},\text{massive}}(a) = \sum_{\kappa=(h,Q)} \frac{(-1)^Q}{2} N_{\kappa}^{(a)} \# \left\{ n \in \mathbb{Z} \mid \Delta = h - \frac{1+Q}{4} + \delta_Q(n) \right\}. \quad (202)$$

The sum in (202) runs over the massive characters extracted below the specified cutoff. The massless descendant coefficient is read from (190). The full DN annulus coefficient is

$$c_{\Delta}^{A,\text{DN}}(a) = c_{\Delta}^{A,\text{DN},\text{massless}}(a) + c_{\Delta}^{A,\text{DN},\text{massive}}(a). \quad (203)$$

The complete c_{Δ} in (185) is obtained from (203) by adding the instanton-instanton annulus contribution, which vanishes representation by representation under (40) in the present assumptions, and the Mobius coefficients computed from the orientifold action.

13.3 Direct Extraction of the Parent DN Annulus

We now carry out the finite extraction, rather than only describing it. This subsection uses Assumption 4: the parent ED3-D7 internal boundary condition is the same on the two ends, so the parent internal oscillator generating function is (53). The computation is independent of floating point numerics; it is a finite manipulation of formal q -series.

The purpose of the extraction is to compute $Z_{A,\text{DN}}^{(a),\text{det}}$, the determinant annulus with one ED3 boundary of fractional label a and one D7 boundary summed over the D7 stack. The final answer will depend on the D7 multiplicities only through the imbalance $\mathcal{D}_a = 2N_a - N_{a-1} - N_{a+1}$.

The character normalization used for the extraction is

$$\hat{\chi}_{h,Q}(u; q) = q^{3/8} \frac{q^{h-(1+Q)/4}}{\eta(\tau)^3} \vartheta_{00}(z|\tau) \vartheta_{Q0}(2z|2\tau), \quad u = e^{2\pi i z}. \quad (204)$$

The factor $q^{3/8}$ removes the internal $c = 9$ vacuum energy, so $\hat{\chi}_{h,Q}$ begins at q^h . With the same normalization,

$$\hat{\chi}_{\text{vac}} = \hat{\chi}_{0,0} - \hat{\chi}_{1/2,1}, \quad \hat{\chi}_{\pm} = \pm \frac{1}{2} q^{3/8} [f_{3,1}(z, \tau) - f_{3,-1}(z, \tau)] + \frac{1}{2} \hat{\chi}_{1/2,1}. \quad (205)$$

Here $f_{3,\pm 1}$ are the APS functions defined by

$$f_{k,Q}(z, \tau) = \eta(\tau)^{-1} \sum_{m \in \mathbb{Z} + Q/k} q^{km^2/2} u^{km}.$$

Equation (205) begins as $\hat{\chi}_{\text{vac}} = 1 + O(q)$, $\hat{\chi}_{+} = u q^{1/2} + O(q)$, and $\hat{\chi}_{-} = u^{-1} q^{1/2} + O(q)$. These are precisely the three massless rows in (51).

For the finite extraction it is convenient to store every exponent as an integer half-unit e , meaning $q^{e/2}$, and every coefficient as a rational number. One first expands $\mathcal{P}_{\text{NS}}(x, u, +1; q)$, then subtracts

$$\hat{\chi}_{\text{vac}} \quad \text{from the } r = 0 \text{ sector,} \quad 3\hat{\chi}_{+} \quad \text{from the } r = 1 \text{ sector,} \quad 3\hat{\chi}_{-} \quad \text{from the } r = 2 \text{ sector,}$$

and finally applies the triangular extraction (197). The first massive multiplicities are

h	Q	$d_{h,Q;0}^{IS}$	$d_{h,Q;1}^{IS}$	$d_{h,Q;2}^{IS}$	$8 \sum_r d_{h,Q;r}^{IS}$
1	0	8	0	0	64
3/2	1	0	6	6	96
2	0	9	15	15	312
5/2	1	20	18	18	448
3	0	56	48	48	1216
7/2	1	70	69	69	1664
4	0	177	168	168	4104

The last column is the total coefficient $N_\kappa^{(a)}$ for the symmetric tadpole solution $N_0 = N_1 = N_2 = 8$; it is independent of a because (50) applies.

Substituting these extracted multiplicities into (202) and adding the exact massless descendant coefficients (194) gives

$$c_\Delta^{A,\text{DN}}(a) = 0, \quad \Delta = 1, 2, \dots, 20, \quad N_0 = N_1 = N_2 = 8. \quad (206)$$

This is not a numerical approximation: the vanishing in (206) is exact coefficient cancellation through the displayed cutoff. The all-order theta-series form of this cancellation is derived next. The cancellation is a determinant supertrace statement only. It does not remove the charged $L_0 = 0$ vertex spaces found in (89).

For use with a non-symmetric crosscap vector, the same extraction gives a compact linear form, and in fact the finite table is the beginning of an exact theta series. Define the A_2 root-lattice theta function by

$$\Theta_{A_2}(q) = \sum_{p,s \in \mathbb{Z}} q^{p^2+ps+s^2} = 1 + \sum_{\Delta=1}^{\infty} r_{A_2}(\Delta) q^\Delta. \quad (207)$$

Here p and s are lattice summation variables, and $r_{A_2}(\Delta)$ is the number of integer pairs (p, s) with $p^2 + ps + s^2 = \Delta$. Equivalently,

$$r_{A_2}(\Delta) = 6 \sum_{d|\Delta} \chi_3(d), \quad \chi_3(d) = \begin{cases} 1, & d \equiv 1 \pmod{3}, \\ -1, & d \equiv 2 \pmod{3}, \\ 0, & d \equiv 0 \pmod{3}. \end{cases} \quad (208)$$

This is the standard divisor formula for the norm form of the Eisenstein-integer lattice A_2 .

Let $\mathcal{F}_\ell(q)$ denote the complete positive-energy DN determinant series obtained by applying the APS DN block to the ℓ -twined parent character expansion in (56), after the $L_0 = 0$ row constants in (171) have been removed. Concretely, this means applying (41) to every massive character coefficient and applying (163) and (166) to the three massless rows, then subtracting the row constants in (172)–(173). Performing this operation on the q -product (53) and using the Jacobi triple-product identity gives the exact result

$$\mathcal{F}_0(q) = 0, \quad \mathcal{F}_1(q) = \mathcal{F}_2(q) = \frac{9}{4} (\Theta_{A_2}(q) - 1). \quad (209)$$

The first equality is the untwined BPS cancellation. The two nontrivial twined sectors are equal because the g - and g^2 -twined traces are complex conjugate before the final real supertrace. Equation (209) is a statement about the determinant supertrace; it is not a statement that the charged zero-mode vertex spaces vanish.

Fourier transforming the three twined determinant series gives the ordered fractional ED3-D7 series

$$F_r(q) = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-r\ell} \mathcal{F}_\ell(q) = \begin{cases} \frac{3}{2} (\Theta_{A_2}(q) - 1), & r = 0, \\ -\frac{3}{4} (\Theta_{A_2}(q) - 1), & r = 1, 2, \end{cases} \quad (210)$$

where $r = a - m \in \mathbb{Z}/3\mathbb{Z}$ is the fractional charge sector selected by an ED3 label a and a D7 label m . Define the local twisted D7 imbalance seen by the instanton label a by

$$\mathcal{D}_a = 2N_a - N_{a-1} - N_{a+1}, \quad a \in \mathbb{Z}/3\mathbb{Z}. \quad (211)$$

Summing over the D7 stack therefore gives the all-order DN determinant series

$$Z_{A,\text{DN}}^{(a),\text{det}}(t) = \sum_{m=0}^2 N_m F_{a-m}(q) = \frac{3}{4} \mathcal{D}_a (\Theta_{A_2}(q) - 1), \quad q = e^{-2\pi t}. \quad (212)$$

Thus the exact DN annulus coefficients are

$$c_{\Delta}^{A,\text{DN}}(a) = \gamma_{\Delta} \mathcal{D}_a, \quad \gamma_{\Delta} = \frac{3}{4} r_{A_2}(\Delta), \quad \Delta \in \mathbb{Z}_{>0}, \quad (213)$$

and $c_{\Delta}^{A,\text{DN}}(a) = 0$ for half-integer Δ . The first nonzero γ_{Δ} are

Δ	1	3	4	7	9	12	13	16	19
γ_{Δ}	$\frac{9}{2}$	$\frac{9}{2}$	$\frac{9}{2}$	9	$\frac{9}{2}$	$\frac{9}{2}$	9	$\frac{9}{2}$	9

All other integer coefficients in $0 < \Delta \leq 20$ vanish because $r_{A_2}(\Delta) = 0$ for those Δ . For example, at $a = 0$,

$$c_{\Delta}^{A,\text{DN}}(0) = \gamma_{\Delta} (2N_0 - N_1 - N_2).$$

Using the tadpole inverse transform (25), the imbalance is equivalently

$$\mathcal{D}_a = -\sqrt{3} \left(C_1 \omega^{-a} + C_2 \omega^{-2a} \right). \quad (214)$$

Thus the symmetric annulus cancellation is the special case $C_1 = C_2 = 0$. If one studies a different crosscap solution with nonzero twisted components, the ED3-D7 annulus contribution is obtained by inserting (214) into (213). A standalone K_0 for that different BCFT would also require its Mobius trace and the charged zero-mode test.

The corresponding large- t contribution to the determinant integral is therefore also explicit:

$$\mathcal{L}_{\geq t_0}^{A,\text{DN}}(a; \Delta_{\text{max}}) = \frac{3}{8} \mathcal{D}_a \sum_{\Delta=1}^{\Delta_{\text{max}}} r_{A_2}(\Delta) E_1(2\pi \Delta t_0) + R_A(t_0, \Delta_{\text{max}}), \quad (215)$$

where E_1 is defined in (186), and R_A is the omitted $\Delta > \Delta_{\text{max}}$ tail.

The same closed form also fixes the small- t behavior of the annulus exactly. Poisson resummation on the A_2 lattice gives

$$\Theta_{A_2}(e^{-2\pi t}) = \frac{1}{\sqrt{3}t} \Theta_{A_2}(e^{-2\pi/(3t)}). \quad (216)$$

Hence, for $C_A^{\text{DN}}(t) = Z_{A,\text{DN}}^{(a),\text{det}}(t)$,

$$C_A^{\text{DN}}(t) = \frac{\sqrt{3} \mathcal{D}_a}{4t} - \frac{3 \mathcal{D}_a}{4} + O\left(t^{-1} e^{-2\pi/(3t)}\right), \quad t \rightarrow 0^+. \quad (217)$$

This is not optional normalization data: if $\mathcal{D}_a \neq 0$, the instanton-crosscap Mobius integrand must contain the opposite twisted massless-exchange coefficient so that the closed-channel tadpole cancellation seen by the instanton is satisfied. In the notation of (214), the annulus leading term is

$$\frac{\sqrt{3} \mathcal{D}_a}{4t} = -\frac{3}{4t} \left(C_1 \omega^{-a} + C_2 \omega^{-2a} \right). \quad (218)$$

Thus the non-vacuum Mobius trace is constrained in the closed channel: its twisted part must cancel (218) with the APS Mobius cutoff relation in (151). Let μ_a be the coefficient of t^{-1} in the twisted part of the instanton-crosscap Mobius integrand:

$$Z_{M,\text{tw}}^{(a)}(t) = \frac{\mu_a}{t} + O(1), \quad t \rightarrow 0^+. \quad (219)$$

The power-divergent part of the APS exponent (183) is then

$$\left(\frac{\sqrt{3}\mathcal{D}_a}{8} + 2\mu_a \right) \frac{1}{\varepsilon'} + O(\log \varepsilon'). \quad (220)$$

The factor $2\mu_a/\varepsilon'$ comes from integrating the Mobius strip from $\varepsilon'/4$, not from ε' . Therefore tadpole cancellation in this normalization requires

$$\mu_a = -\frac{\sqrt{3}}{16} \mathcal{D}_a = \frac{3}{16} (C_1 \omega^{-a} + C_2 \omega^{-2a}). \quad (221)$$

Equation (221) is a bootstrap equation for the same signed traces $\mathfrak{m}_{h,Q}$ that appear in the open-channel Mobius expansion (144). It fixes a closed-channel moment of the non-vacuum Mobius trace; it is not obtained by multiplying an open-channel character by an O-plane charge.

For later numerical assembly, the annulus small-end finite part can be written without an unstable integral. Define

$$\text{FP}_{0 < t < t_0}^{A,\text{DN}} = \lim_{\epsilon \rightarrow 0^+} \left[\int_{\epsilon}^{t_0} \frac{dt}{2t} C_A^{\text{DN}}(t) - \frac{\sqrt{3}\mathcal{D}_a}{8\epsilon} - \frac{3\mathcal{D}_a}{8} \log \epsilon \right]. \quad (222)$$

Using (216), this is

$$\text{FP}_{0 < t < t_0}^{A,\text{DN}} = \frac{3\sqrt{3}\mathcal{D}_a}{16\pi} \sum_{\Delta=1}^{\infty} \frac{r_{A_2}(\Delta)}{\Delta} e^{-2\pi\Delta/(3t_0)} - \frac{\sqrt{3}\mathcal{D}_a}{8t_0} - \frac{3\mathcal{D}_a}{8} \log t_0. \quad (223)$$

The split point t_0 is only a numerical device. Combining the small-end finite part with the large- t series gives a single annulus finite part

$$\mathcal{J}_{A,\text{DN}}(a) = \lim_{\epsilon \rightarrow 0^+} \left[\int_{\epsilon}^{\infty} \frac{dt}{2t} C_A^{\text{DN}}(t) - \frac{\sqrt{3}\mathcal{D}_a}{8\epsilon} - \frac{3\mathcal{D}_a}{8} \log \epsilon \right]. \quad (224)$$

Here $C_A^{\text{DN}}(t)$ is the exact annulus determinant integrand in (217), and the subtractions are precisely its small- t power and logarithmic terms. Equivalently, for any $t_0 > 0$,

$$\begin{aligned} \mathcal{J}_{A,\text{DN}}(a) &= \frac{3\sqrt{3}\mathcal{D}_a}{16\pi} \sum_{\Delta=1}^{\infty} \frac{r_{A_2}(\Delta)}{\Delta} e^{-2\pi\Delta/(3t_0)} - \frac{\sqrt{3}\mathcal{D}_a}{8t_0} - \frac{3\mathcal{D}_a}{8} \log t_0 \\ &\quad + \frac{3}{8} \mathcal{D}_a \sum_{\Delta=1}^{\infty} r_{A_2}(\Delta) E_1(2\pi\Delta t_0). \end{aligned} \quad (225)$$

The right hand side is independent of t_0 when the sums are not truncated. There is also a closed analytic form. Let

$$\zeta_{A_2}(s) = \sum_{(p,u) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{1}{(p^2 + pu + u^2)^s} = 6 \zeta(s) L(s, \chi_3), \quad (226)$$

where $\zeta(s)$ is the Riemann zeta function and

$$L(s, \chi_3) = \sum_{n=1}^{\infty} \chi_3(n) n^{-s}$$

is the Dirichlet L -function of the same character χ_3 used in (208). The Mellin transform gives

$$\int_0^{\infty} t^{s-1} \left(\Theta_{A_2}(e^{-2\pi t}) - 1 \right) dt = (2\pi)^{-s} \Gamma(s) \zeta_{A_2}(s).$$

Taking the constant term at $s = 0$, using $\zeta_{A_2}(0) = -1$, $L(0, \chi_3) = 1/3$, and $\zeta'(0) = -\frac{1}{2} \log(2\pi)$, gives the exact finite part

$$\mathcal{J}_{A, \text{DN}}(a) = \frac{3\mathcal{D}_a}{8} (\gamma_E - 3L'(0, \chi_3)), \quad (227)$$

where γ_E is Euler's constant. Since

$$L'(0, \chi_3) = \log \Gamma\left(\frac{1}{3}\right) - \log \Gamma\left(\frac{2}{3}\right) - \frac{1}{3} \log 3,$$

(227) is an exact number in this normalization. Per unit $\mathcal{D}_a$,

$$\mathcal{J}_{A, \text{DN}}/\mathcal{D}_a = -0.13911868566415798107456612442984609970. \quad (228)$$

Thus, under the symmetric local tadpole assumption, the annulus contribution to the positive-energy q -series has been computed to all orders: it vanishes because $\mathcal{D}_a = 0$. The Mobius coefficient package $\tilde{n}_{h, Q, \sigma}$ is computed for the folded charge-conjugating branch in (249).

13.4 Mobius Coefficients for the q -series

The Mobius formula (144) is naturally written with $\hat{q} = -q$, where $q = e^{-2\pi t}$. To combine it with the annulus coefficients in (185), first rewrite it as a real q -series. With the branch convention in (144),

$$e^{-i\pi(h-Q/2)} \hat{q}^{h-(1+Q)/4} = e^{i\pi(Q-1)/4} q^{h-(1+Q)/4}. \quad (229)$$

Moreover

$$\vartheta_{10}(2\hat{\tau}) = e^{i\pi/4} q^{1/4} \sum_{n \in \mathbb{Z}} q^{n(n+1)}, \quad \vartheta_{00}(2\hat{\tau}) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2}. \quad (230)$$

Substituting (229) and (230) into (144) gives the exact open-channel expansion

$$\begin{aligned} Z_M(t) = & - \sum_{h, \sigma} \sigma \tilde{n}_{h, 0, \sigma} \sum_{n \in \mathbb{Z}} q^{h+n(n+1)} \\ & + \sum_{h, \sigma} \sigma \tilde{n}_{h, 1, \sigma} \sum_{n \in \mathbb{Z}} (-1)^n q^{h-1/2+n^2}. \end{aligned} \quad (231)$$

The first sum is the $Q = 0$ part and the second sum is the $Q = 1$ part. The coefficients $\tilde{n}_{h, Q, \sigma}$ are the Mobius character coefficients defined in (144); they are not determined by the D7 multiplicities N_m .

For $\Delta > 0$, the Mobius contribution to the coefficient in (185) is therefore

$$\begin{aligned} c_{\Delta}^M = & - \sum_{h, \sigma} \sigma \tilde{n}_{h, 0, \sigma} \# \{n \in \mathbb{Z} \mid \Delta = h + n(n+1)\} \\ & + \sum_{h, \sigma} \sigma \tilde{n}_{h, 1, \sigma} \sum_{\substack{n \in \mathbb{Z} \\ \Delta = h-1/2+n^2}} (-1)^n. \end{aligned} \quad (232)$$

The $\Delta = 0$ coefficient is not used in (185): it is removed by the zero-mode subtraction in (184). For the universal $O(1)$ vacuum data (146),

$$Z_M^{\text{vac}}(t) - 3 = \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0, -1}} q^{n(n+1)} + \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} (-1)^n q^{n^2}. \quad (233)$$

This checks the subtraction convention: Z_M^{vac} has three constant contributions, two from the $Q = 0$ theta series and one from the $Q = 1$ theta series, and these are exactly the APS constant removed in (184).

13.5 Universal Mobius Vacuum Finite Part

The universal vacuum series in (233) is now fully integrated. Write

$$q = e^{-2\pi t}, \quad Z_M^{\text{vac}}(t) - 3 = 2 \sum_{k=1}^{\infty} q^{k(k+1)} + 2 \sum_{n=1}^{\infty} (-1)^n q^{n^2}. \quad (234)$$

With an open-channel cutoff δ in the variable t , the corresponding integral is

$$I_M^{\text{vac}}(\delta) = \int_{\delta}^{\infty} \frac{dt}{2t} (Z_M^{\text{vac}}(t) - 3) = \sum_{k=1}^{\infty} E_1(2\pi k(k+1)\delta) + \sum_{n=1}^{\infty} (-1)^n E_1(2\pi n^2\delta). \quad (235)$$

Here $E_1(x) = \int_x^{\infty} e^{-u} du/u$. The small- δ expansion is

$$I_M^{\text{vac}}(\delta) = \frac{1}{\sqrt{2\delta}} + \frac{3}{2} \log \delta + \frac{3}{2} \gamma_E + \frac{3}{2} \log 2 - \frac{1}{2} \log \pi + o(1). \quad (236)$$

The two pieces in (235) give this result as follows:

$$\text{FP}_{\delta \rightarrow 0} \left[\sum_{k=1}^{\infty} E_1(2\pi k(k+1)\delta) - \frac{1}{\sqrt{2\delta}} - \log \delta \right] = \gamma_E, \quad (237)$$

$$\text{FP}_{\delta \rightarrow 0} \left[\sum_{n=1}^{\infty} (-1)^n E_1(2\pi n^2\delta) - \frac{1}{2} \log \delta \right] = \frac{1}{2} \gamma_E + \frac{3}{2} \log 2 - \frac{1}{2} \log \pi. \quad (238)$$

The second identity follows directly from $\sum_{n \geq 1} (-1)^n n^{-2s} = -\eta(2s)$, where $\eta(s) = (1 - 2^{1-s})\zeta(s)$. The first is the Mellin finite part of the heat kernel $\sum_{k \geq 1} \exp[-2\pi t k(k+1)]$; equivalently, it follows by applying Euler–Maclaurin to $E_1(2\pi \delta x(x+1))$.

The APS regulator uses the common closed-channel cutoff ε' , with the Mobius lower limit $\delta = \varepsilon'/4$. Therefore the Mobius vacuum finite part that enters the same subtraction scheme as the annulus is shifted by the cutoff relation:

$$\mathcal{F}_M^{\text{vac}} = \frac{3}{2} \gamma_E - \frac{3}{2} \log 2 - \frac{1}{2} \log \pi = -0.74626221641231876028779372274019056137. \quad (239)$$

This is a completed contribution to $\mathcal{K}_0 = \log K_0$. It is independent of the fractional label a and of the D7 multiplicities. It is only the vacuum row; the non-vacuum Mobius traces of the folded one-node branch are computed in the next subsection.

13.6 Folded Charge-Conjugating Mobius Trace

For the O7 lift (21), the direct-channel Mobius operator conjugates the internal $U(1)$ charge. Therefore its trace is not an ordinary twined character with a spectator u -charge: it is a folded trace over charge-conjugate oscillator pairs. In the NS sector the consistent folded branch has fermion-pair square $\eta_F = -1$. The branch $\eta_F = +1$ gives an exponentially growing small- t transform and is not a closed-string tree-channel factorization of the O7 BCFT.

This subsection concerns only the neutral ED3-ED3 sector. The D7 multiplicities N_m do not enter the folded Mobius trace; they enter separately through the DN annulus and the zero-mode test.

Let

$$\Xi_0(q) = \prod_{n=1}^{\infty} \frac{1 + q^{2n-1}}{1 - q^{2n}}. \quad (240)$$

This is the folded twining character of a generic massive $Q = 0$ module. The generic massive $Q = 1$ folded twining character vanishes, because the charge-conjugating spectral-flow lattice has no fixed $Q = 1$ sector in this basis. The raw charge-conjugating oscillator trace is

$$\mathcal{R}_{\Theta}(q) = \prod_{n=1}^{\infty} \frac{(1 + q^{2n-1})^3}{(1 - q^{2n})^3}. \quad (241)$$

This is already the oscillator factor that appears in every orbifold-twined Mobius trace of the one-node branch. To see this directly, let

$$\Theta_{\ell} = \Omega_{\sigma} g^{\ell}, \quad \ell = 0, 1, 2, \quad (242)$$

be the Mobius insertion before the $\mathbb{Z}_3$ average. On the cover fields, Θ_{ℓ} acts on a conjugate bosonic oscillator pair with phases $-\omega^{\ell}$ and $-\omega^{-\ell}$. Their product is one. Therefore a single complex bosonic pair at oscillator level n contributes

$$\text{Tr}_{\text{bos,pair}}(\Theta_{\ell} q^{nN}) = \frac{1}{1 - q^{2n}}, \quad (243)$$

where N is the occupation-number operator for that pair. In the consistent NS folded branch the corresponding fermion-pair square is $\eta_F = -1$, so a conjugate fermionic pair at $r \in \mathbb{Z}_{\geq 0} + \frac{1}{2}$ contributes

$$\text{Tr}_{\text{ferm,pair}}^{\text{NS}}(\Theta_{\ell} q^{rN}) = 1 + q^{2r}. \quad (244)$$

Equations (243) and (244) are independent of ℓ . For three complex internal directions their product is exactly $\mathcal{R}_{\Theta}(q)$ in (241).

It remains to include the one-dimensional Chan-Paton phase. If the orientifold lift is the character lift s in (18), then the Θ_{ℓ} insertion on the fractional label a carries the phase

$$\chi_{a,s}(\ell) = \omega^{(2a-s)\ell}. \quad (245)$$

The orbifold-averaged raw trace is consequently

$$\mathcal{R}_{\Theta}^{(a,s)}(q) = \frac{1}{3} \sum_{\ell=0}^2 \chi_{a,s}(\ell) \mathcal{R}_{\Theta}(q) = \begin{cases} \mathcal{R}_{\Theta}(q), & 2a = s \pmod{3}, \\ 0, & 2a \neq s \pmod{3}. \end{cases} \quad (246)$$

The one-node O(1) instanton is precisely the invariant case $2a = s$. Thus (241) is not the $\ell = 0$ term alone; it is the full orbifold-averaged instanton-instanton Mobius oscillator trace for the invariant

fractional label. Labels with $2a \neq s$ are exchanged with an image label and require a brane-image system rather than a one-node $O(1)$ trace. After subtracting the vacuum row and decomposing into shifted copies of Ξ_0 , write

$$\frac{\mathcal{R}_\Theta(q)}{\Xi_0(q)} - 1 = \sum_{N=1}^{\infty} \rho_N q^N. \quad (247)$$

The APS real- q convention in (231) then gives

$$\mathbf{m}_{N,0}^{\text{fold}} = (-1)^N \rho_N, \quad \mathbf{m}_{h,1}^{\text{fold}} = 0, \quad (248)$$

with $N \in \mathbb{Z}_{\geq 1}$. Equivalently,

$$\sum_{N=1}^{\infty} \mathbf{m}_{N,0}^{\text{fold}} q^N = G(q) - 1, \quad G(q) = \prod_{n=1}^{\infty} \frac{(1 - q^{2n-1})^2}{(1 - q^{2n})^2}. \quad (249)$$

The first values are

N	1	2	3	4	5	6	7	8
$\mathbf{m}_{N,0}^{\text{fold}}$	-2	3	-6	11	-18	28	-44	69

(250)

For any physical instanton-instanton Mobius trace, let $D_{h,Q}^{II}$ be the dimension of the charge-zero multiplicity space of the massive representation (h, Q) before the orientifold sign is diagonalized. The signed trace must obey

$$\mathbf{m}_{h,Q} \in \mathbb{Z}, \quad |\mathbf{m}_{h,Q}| \leq D_{h,Q}^{II}, \quad \mathbf{m}_{h,Q} \equiv D_{h,Q}^{II} \pmod{2}. \quad (251)$$

They obey the integrality, parity, and absolute-value bounds in (251); for example $|\mathbf{m}_{1,0}^{\text{fold}}| = 2 \leq D_{1,0}^{II} = 8$ and $\mathbf{m}_{1,0}^{\text{fold}} \equiv D_{1,0}^{II} \pmod{2}$.

Define

$$T(q) = \sum_{n \in \mathbb{Z}} q^{n(n+1)}, \quad S(q) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2}, \quad q = e^{-2\pi t}. \quad (252)$$

The full folded Mobius determinant integrand, including the universal vacuum row and the massive folded trace, is

$$C_M^{\text{fold}}(t) = T(q) + S(q) - 3 - T(q)(G(q) - 1) = 2T(q) + S(q) - 3 - T(q)G(q). \quad (253)$$

The modular transform of this exact expression is

$$\begin{aligned} C_M^{\text{fold}}(t) &= \sqrt{\frac{2}{t}} e^{\pi t/2} \vartheta_{01}\left(\frac{i}{2t}\right) + \frac{1}{\sqrt{2t}} \vartheta_{10}\left(\frac{i}{2t}\right) - 3 \\ &\quad - 2\sqrt{2t} \vartheta_{01}\left(\frac{i}{2t}\right) \frac{\eta(i/t)^2}{\eta(i/(2t))^4}. \end{aligned} \quad (254)$$

Consequently,

$$C_M^{\text{fold}}(t) = \sqrt{\frac{2}{t}} - 3 + O(\sqrt{t}), \quad t \rightarrow 0^+. \quad (255)$$

The finite part with a cutoff in the open variable t is

$$F_{M,t}^{\text{fold}} = -0.74316012896187184035872169945492327348. \quad (256)$$

This is not the APS finite part. Since the APS formula cuts off the Mobius integral at $\delta = \varepsilon'/4$, the logarithmic term in (255) shifts the finite part by $-\frac{3}{2} \log 4$. The Mobius contribution in the APS convention is therefore

$$\mathcal{F}_M^{\text{fold}} = F_{M,t}^{\text{fold}} - \frac{3}{2} \log 4 = -2.82260167064170776861041806382945297770. \quad (257)$$

The non-vacuum folded contribution alone is

$$\mathcal{F}_{M,\text{nonvac}}^{\text{fold}} = \mathcal{F}_M^{\text{fold}} - \mathcal{F}_M^{\text{vac}} = -2.07633945422938900832262434108926241633. \quad (258)$$

Equations (249) and (257) complete the instanton-instanton Mobius BCFT data for the one-node folded branch.

Combining the pieces, under the present assumptions of no nonzero instanton-instanton annulus determinant,

$$c_\Delta = c_\Delta^{A,\text{DN}}(a) + c_\Delta^M, \quad \Delta > 0. \quad (259)$$

Equation (259) is the coefficient-level object that enters the positive-energy determinant integral. When the D7 imbalance obeys $\mathcal{D}_a = 0$, the annulus term is zero and $c_\Delta = c_\Delta^M$ is fixed by the folded product above. For $\mathcal{D}_a \neq 0$, the annulus term is the A_2 -theta series (212); the next section shows that this produces an uncanceled twisted tadpole in the one-node folded branch, rather than an extra coefficient to tune.

14 Tadpole Compatibility of the Completed BCFT Data

The preceding sections have fixed the two determinant integrands in the one-node folded branch. For $q = e^{-2\pi t}$, the instanton-instanton annulus vanishes and the nonzero positive-energy determinant integrand is

$$C_a^{\text{det}}(t) = C_{A,\text{DN}}^{(a)}(t) + C_M^{\text{fold}}(t) = \frac{3}{4} \mathcal{D}_a (\Theta_{A_2}(q) - 1) + C_M^{\text{fold}}(t), \quad (260)$$

where $C_M^{\text{fold}}(t)$ is the Mobius function in (253) and

$$\mathcal{D}_a = 2N_a - N_{a-1} - N_{a+1}. \quad (261)$$

Here N_m is the D7 multiplicity of fractional label m , and all labels are read in $\mathbb{Z}/3\mathbb{Z}$. Equation (260) is the complete BCFT input for the branch whose one-node Mobius trace is (246).

This comparison is a tadpole test for the same branch. It does not introduce a tunable Mobius coefficient: the folded Mobius trace has already been fixed by the orientifold action on the ED3-ED3 Hilbert space.

The small- t behavior follows from (217) and (255):

$$C_a^{\text{det}}(t) = \frac{\sqrt{3}\mathcal{D}_a}{4t} + \sqrt{\frac{2}{t}} - \frac{3\mathcal{D}_a}{4} - 3 + O(\sqrt{t}) + O\left(t^{-1}e^{-2\pi/(3t)}\right). \quad (262)$$

The $\sqrt{2/t}$ and -3 terms are the universal folded Mobius terms whose APS finite part was evaluated in (257). The term proportional to $\mathcal{D}_a/t$ is different: it is the twisted massless exchange from the ED3-D7 annulus. Since the one-node folded Mobius trace has no t^{-1} term, this branch is tadpole-compatible only if

$$\mathcal{D}_a = 0. \quad (263)$$

Equivalently, with the one-node image constraint $N_{a-1} = N_{a+1}$, the condition is $N_a = N_{a-1} = N_{a+1}$. If $\mathcal{D}_a \neq 0$, the APS exponent contains the uncanceled endpoint terms

$$\frac{\sqrt{3}\mathcal{D}_a}{8\varepsilon'} + \frac{3\mathcal{D}_a}{8} \log \varepsilon', \quad \varepsilon' \rightarrow 0^+, \quad (264)$$

from the annulus integral. Thus a non-symmetric D7 vector is not a partially computed K_0 in this branch. It defines a different tadpole problem, with different crosscap and Chan–Paton data, and must be treated as a different orientifold BCFT before a standalone K_0 integral is meaningful.

15 Status of the K_0 Computation

Write

$$K_0(a) = \exp \mathcal{K}_0(a), \quad (265)$$

where $a \in \mathbb{Z}/3\mathbb{Z}$ is the fractional instanton label. The BCFT data entering the one-node folded branch are now

$$Z_{A,DD}^{\det}(t) = 0, \quad (266)$$

$$Z_{A,DN}^{(a),\det}(t) = \frac{3}{4} \mathcal{D}_a \left(\Theta_{A_2}(e^{-2\pi t}) - 1 \right), \quad \mathcal{D}_a = 2N_a - N_{a-1} - N_{a+1}, \quad (267)$$

$$Z_M^{\det}(t) = C_M^{\text{fold}}(t), \quad (268)$$

with C_M^{fold} defined in (253). The corresponding finite parts are

$$\mathcal{J}_{A,DD}(a) = 0, \quad (269)$$

$$\mathcal{J}_{A,DN}(a) = \mathcal{D}_a J_{A_2}, \quad (270)$$

$$J_{A_2} = \frac{3}{8} (\gamma_E - 3L'(0, \chi_3)), \quad (271)$$

$$J_{A_2} = -0.13911868566415798107456612442984609970, \quad (271)$$

$$\mathcal{F}_M^{\text{fold}} = -2.82260167064170776861041806382945297770. \quad (272)$$

Here γ_E is Euler's constant and $L(s, \chi_3)$ is the Dirichlet L -function defined below (226). The Mobius number in (272) is in the APS cutoff convention; the open- t finite part $F_{M,t}^{\text{fold}}$ in (256) is not the APS exponent.

These formulas give a standalone APS K_0 only after two independent tests are passed. The first is twisted-tadpole compatibility of the branch:

$$\mathcal{D}_a = 0. \quad (273)$$

If $\mathcal{D}_a \neq 0$, the exponent has the uncanceled endpoint behavior displayed in (264); there is no finite standalone K_0 in the one-node folded branch. The second is the absence of non-universal zero modes:

$$\mathcal{Z}_{\text{extra}}^{\text{proj}}(a) = \mathcal{Z}_{\text{NS}}^{\text{proj}}(a) \oplus \mathcal{Z}_{R,\text{extra}}^{\text{proj}}(a) = 0. \quad (274)$$

Here $\mathcal{Z}_{\text{NS}}^{\text{proj}}(a)$ is the projected charged NS $L_0 = 0$ space, and $\mathcal{Z}_{R,\text{extra}}^{\text{proj}}(a)$ is the projected Ramond $L_0 = 0$ space after removing the two universal Goldstinos θ^α .

When both (273) and (274) hold, the determinant exponent is therefore

$$\boxed{\mathcal{K}_0(a) = 0 + \mathcal{D}_a J_{A_2} + \mathcal{F}_M^{\text{fold}} = -2.82260167064170776861041806382945297770} \quad (275)$$

and

$$\boxed{K_0(a) = 0.05945106922997143616905682111809587940.} \quad (276)$$

The three terms in (275) are, in order, the instanton-instanton annulus, the ED3-D7 annulus, and the one-node folded Mobius contribution. The middle term vanishes because $\mathcal{D}_a = 0$ is required by (273).

In the ordinary symmetric local D7 realization studied in this note,

$$N_0 = N_1 = N_2 = 8, \quad \dim \mathcal{Z}_{\text{NS}}^{\text{proj}}(a) = 16, \quad \mathcal{Z}_{\text{extra}}^{\text{proj}}(a) \neq 0. \quad (277)$$

Thus the ordinary model satisfies the determinant tadpole test $\mathcal{D}_a = 0$, but fails the zero-mode test. It has the completed local determinant factor (276); it does not have a standalone APS superpotential prefactor without also integrating or lifting the charged zero modes.

Proposition 3 (Final BCFT outcome). *Under the assumptions of this note, the BCFT data entering the one-node folded $O(1)$ branch are complete and are precisely (266)–(268). Therefore:*

1. *If $\mathcal{D}_a \neq 0$, the annulus produces the uncanceled endpoint terms (264). The one-node folded branch then has no finite standalone APS K_0 .*
2. *If $\mathcal{D}_a = 0$ and $\mathcal{Z}_{\text{extra}}^{\text{proj}}(a) = 0$, the standalone determinant is (275)–(276).*
3. *In the ordinary local OT^- tadpole solution with $N_0 = N_1 = N_2 = 8$, the first condition holds but the second fails because $\dim \mathcal{Z}_{\text{NS}}^{\text{proj}}(a) = 16$. Thus this ordinary model has the completed determinant factor (276), but not a standalone superpotential prefactor K_0 .*

16 Bibliographic Anchors

The normalization formula (183), the universal DN block (41), and the Mobius formula (144) are imported from Alexandrov–Pioline–Sen’s D-instanton formalism. The fractional boundary-state construction (10) and eigenspace rule (11) are imported from the string book’s treatment of D-branes in orbifolds.

References

- [1] D. Alexandrov, B. Pioline, and A. Sen, *D-instanton induced superpotential*, arXiv:2204.02981.
- [2] X. Yin, *String notes*, local reference file `references/stringbook-main/texsource/stringnotes.tex`.